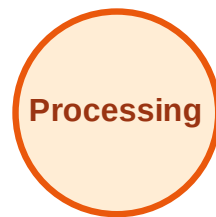


BFS Traversal

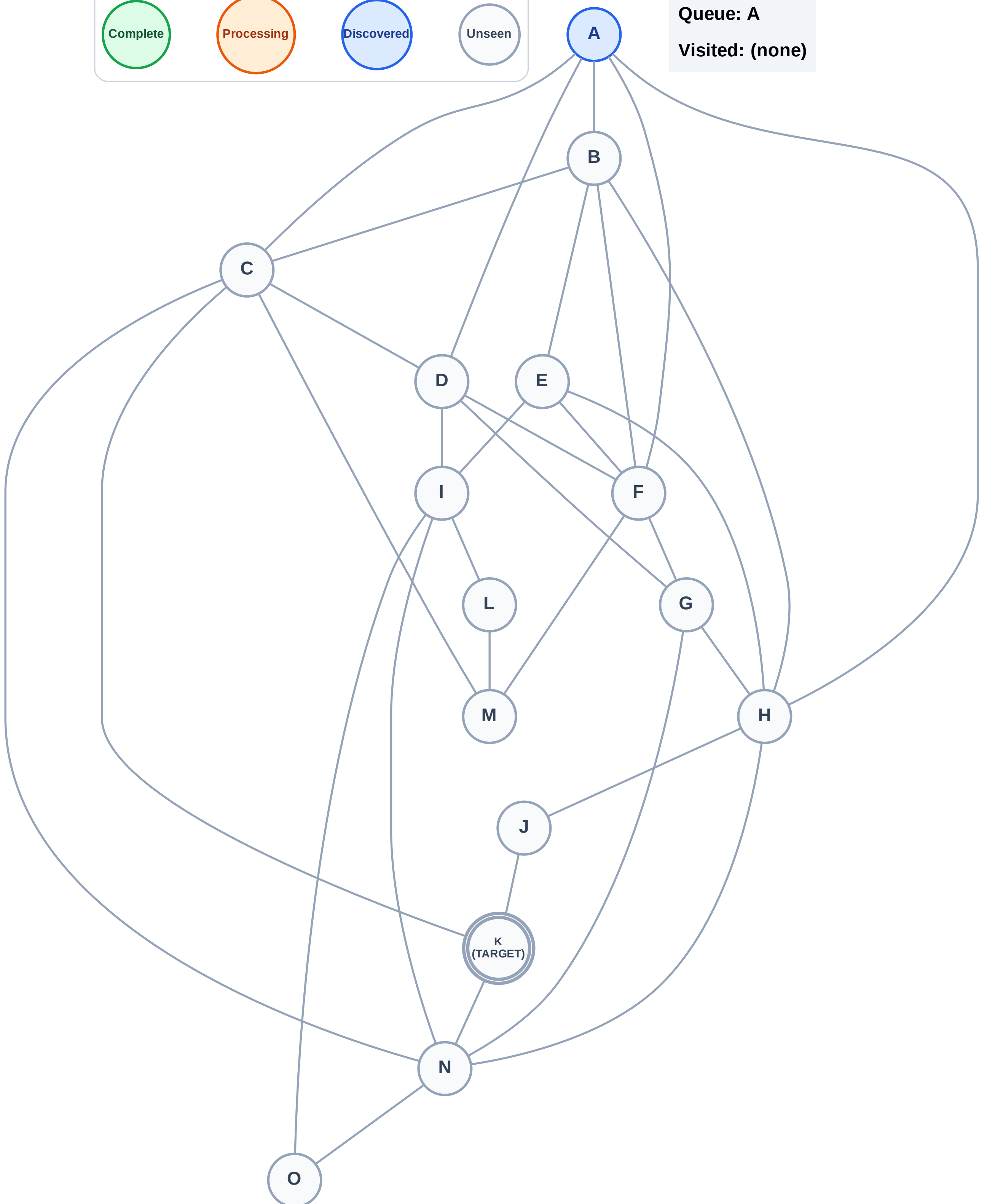
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

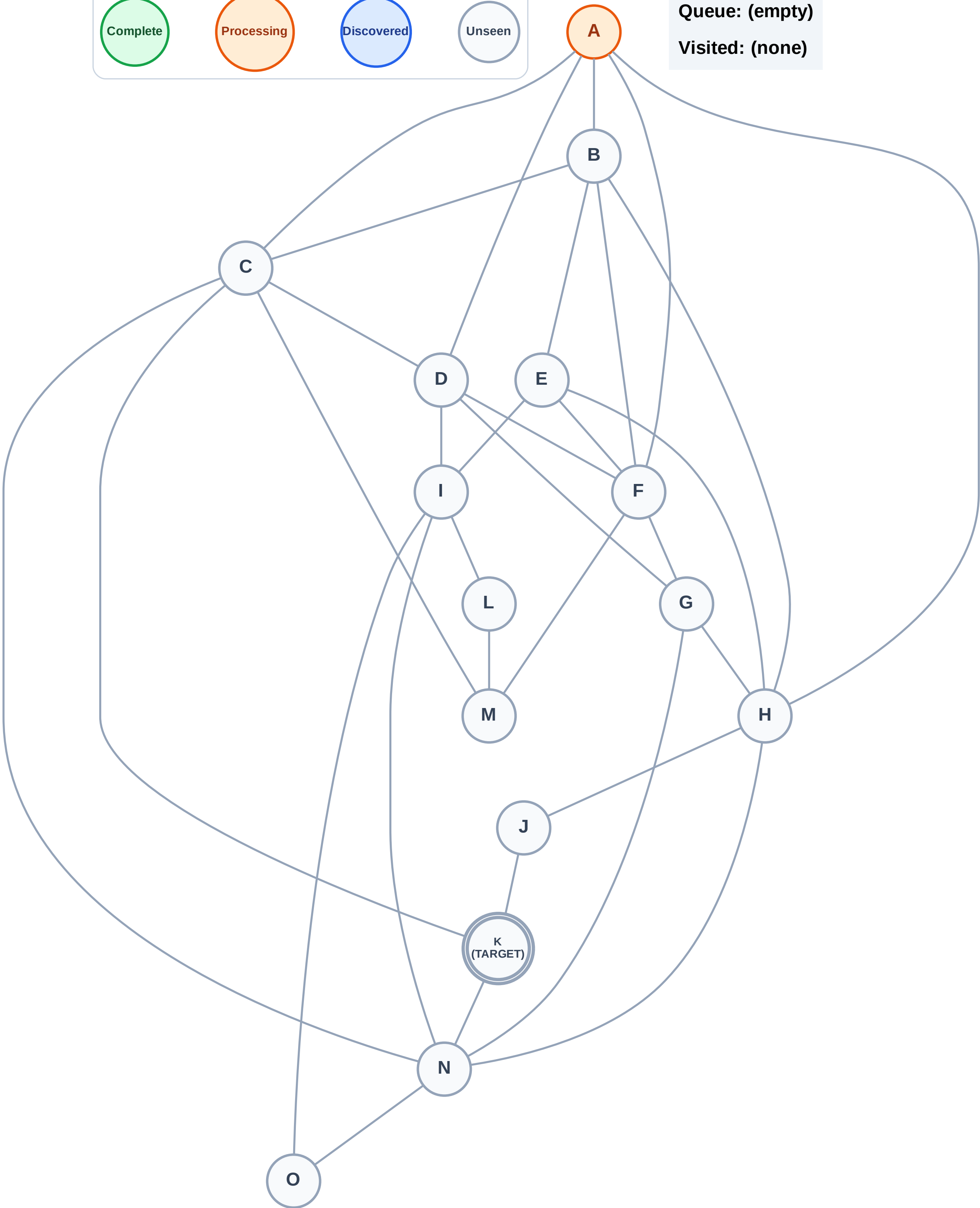
Complete

Processing

Discovered

Unseen

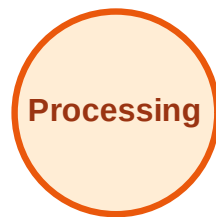
Queue: (empty)
Visited: (none)



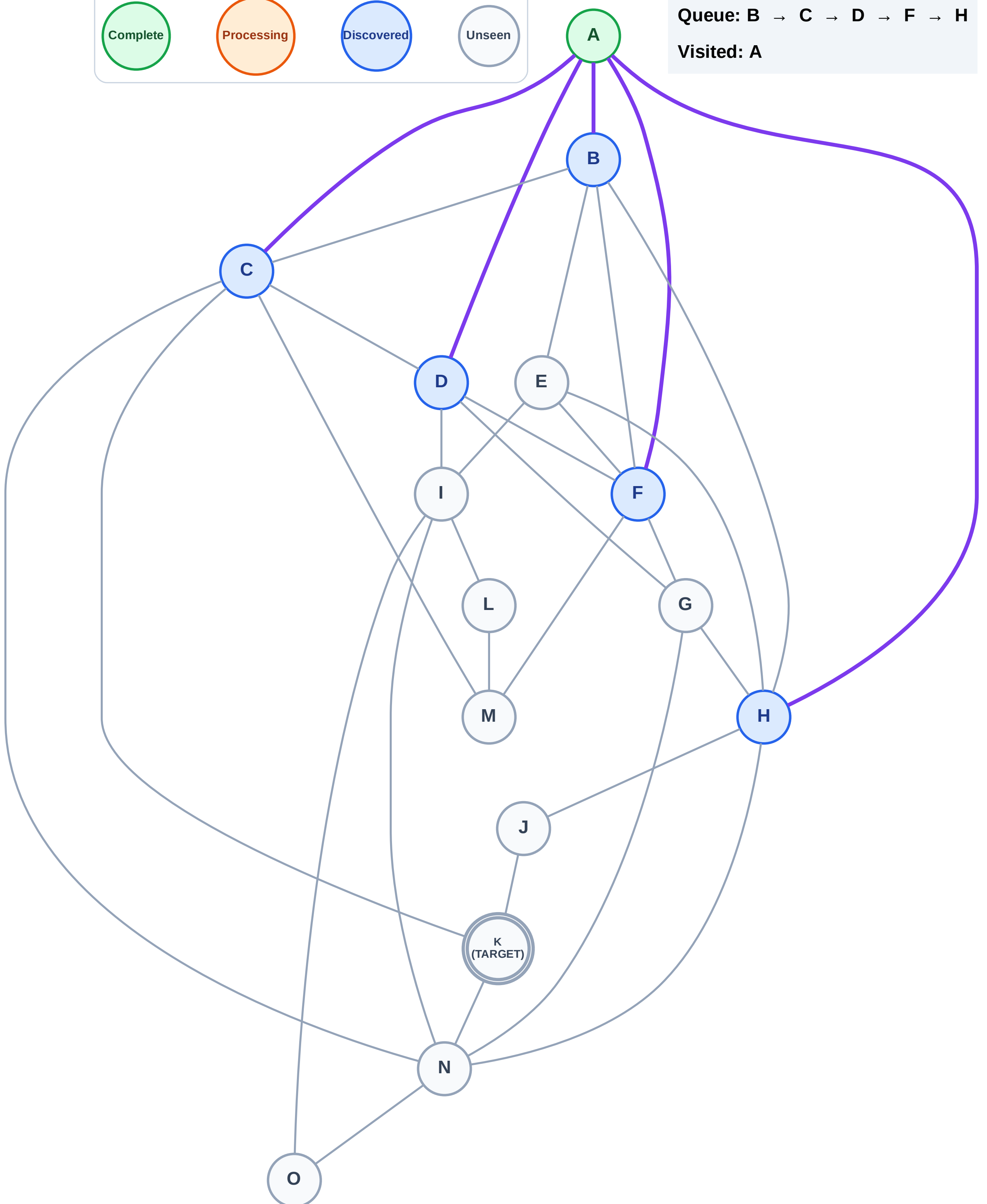
BFS Traversal

Step 3: Discover B, C, D, F, H from A

Legend



Queue: B → C → D → F → H
Visited: A



BFS Traversal

Step 4: Process node B

Legend

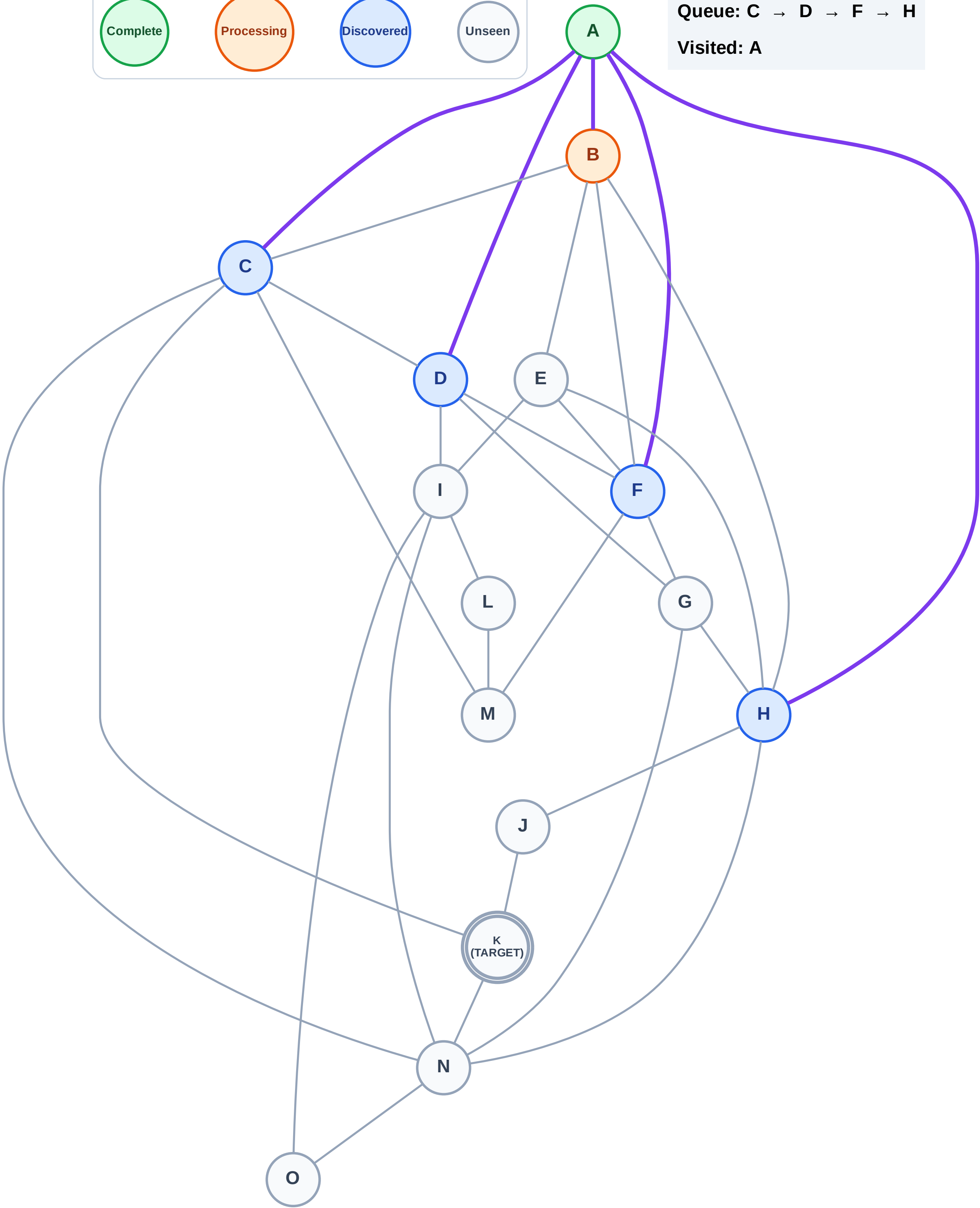
Complete

Processing

Discovered

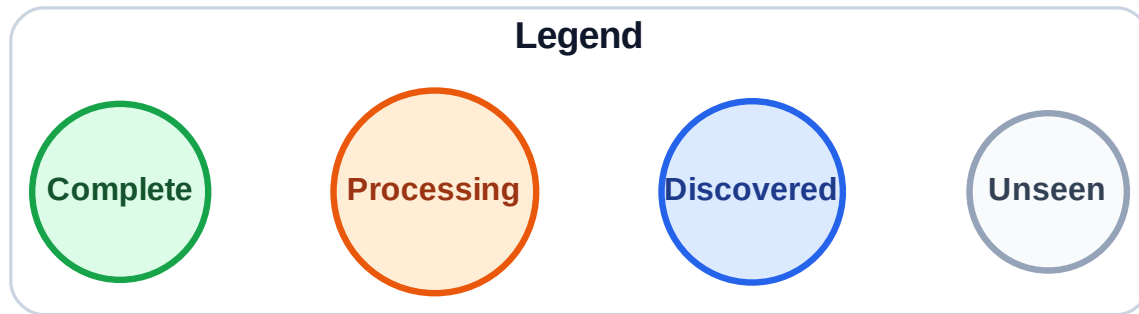
Unseen

Queue: C → D → F → H
Visited: A



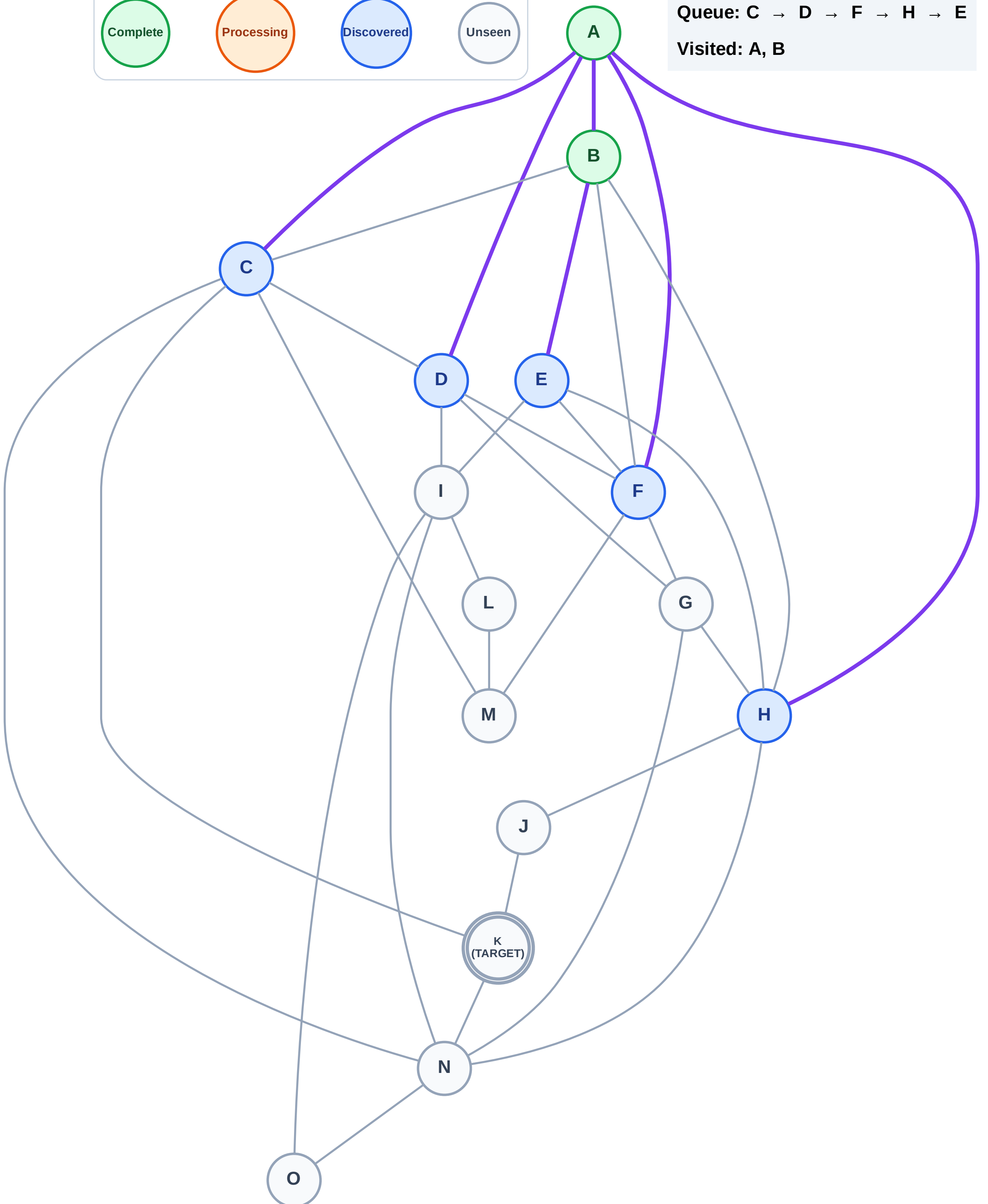
BFS Traversal

Step 5: Discover E from B



Queue: C → D → F → H → E

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

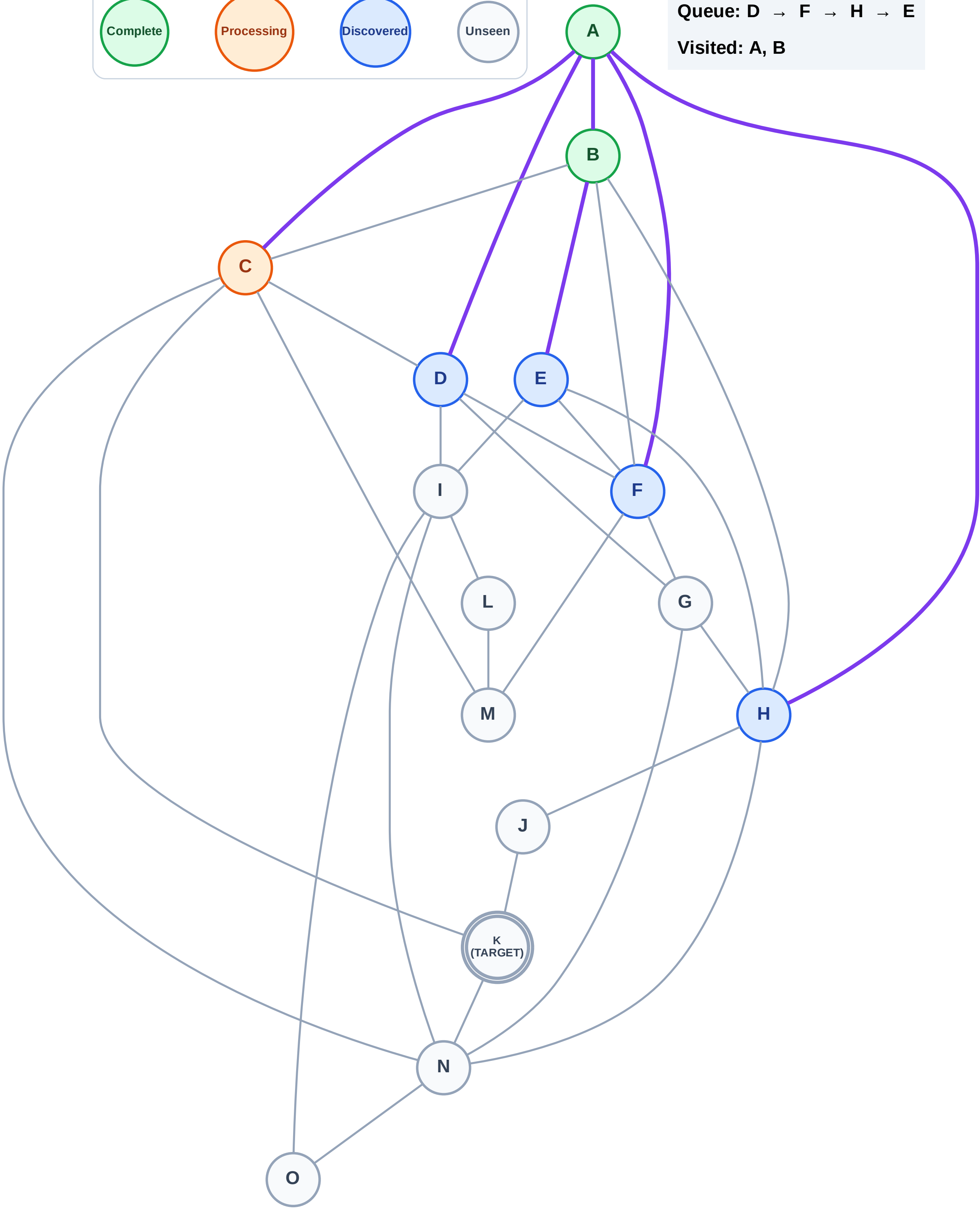
Processing

Discovered

Unseen

Queue: D → F → H → E

Visited: A, B



BFS Traversal

Step 7: Discover K, M, N from C

Legend

Complete

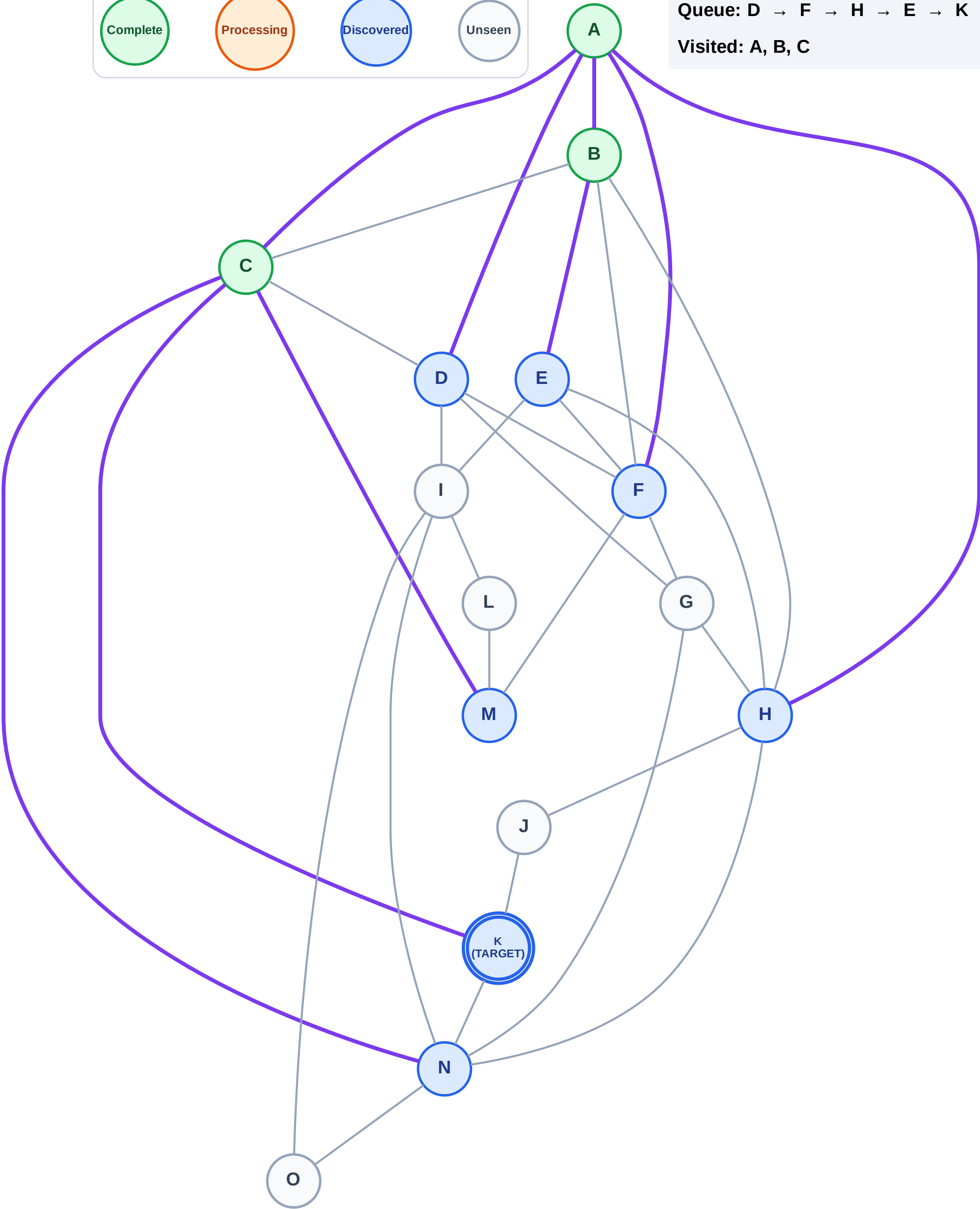
Processing

Discovered

Unseen

Queue: D → F → H → E → K → M → N

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

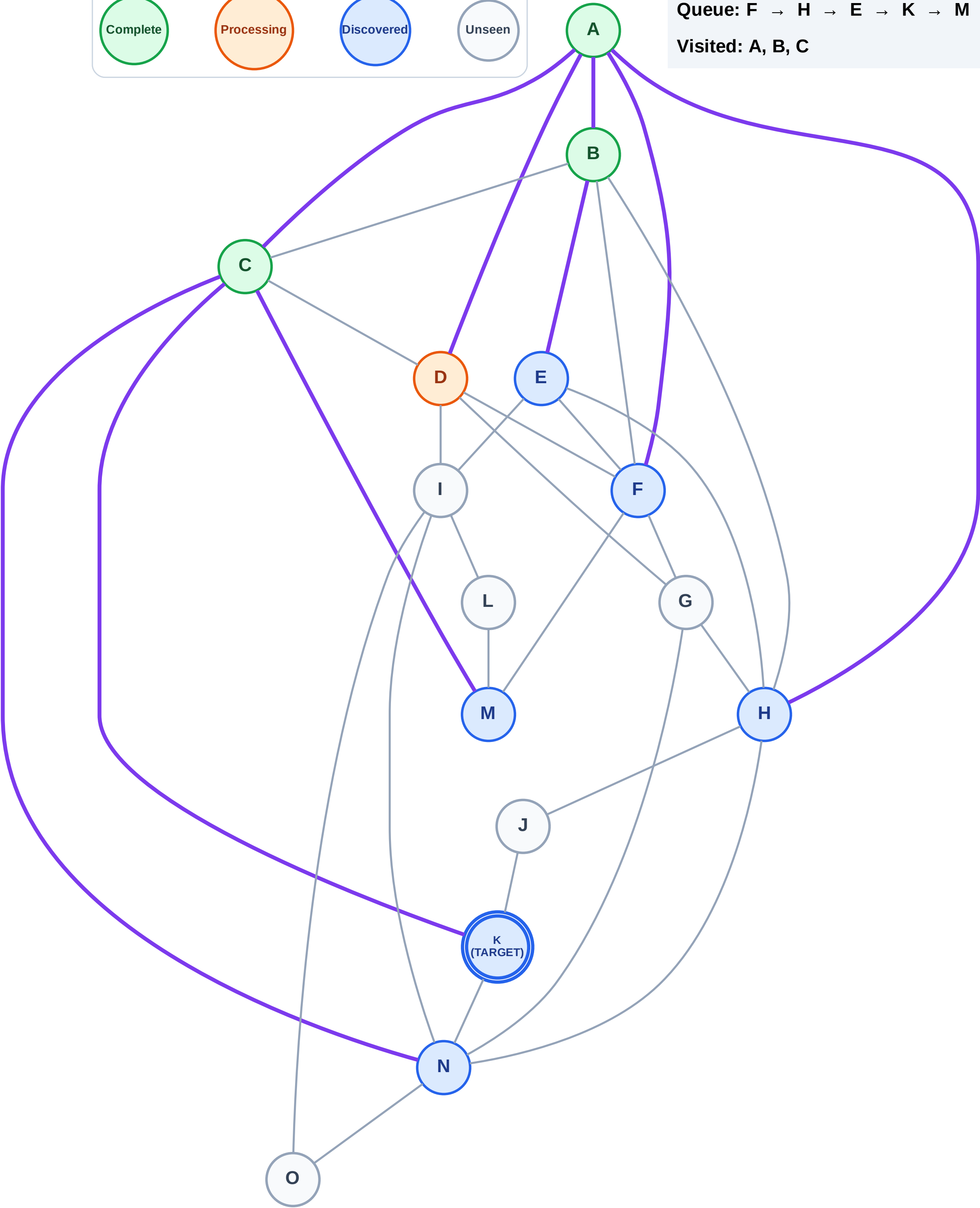
Processing

Discovered

Unseen

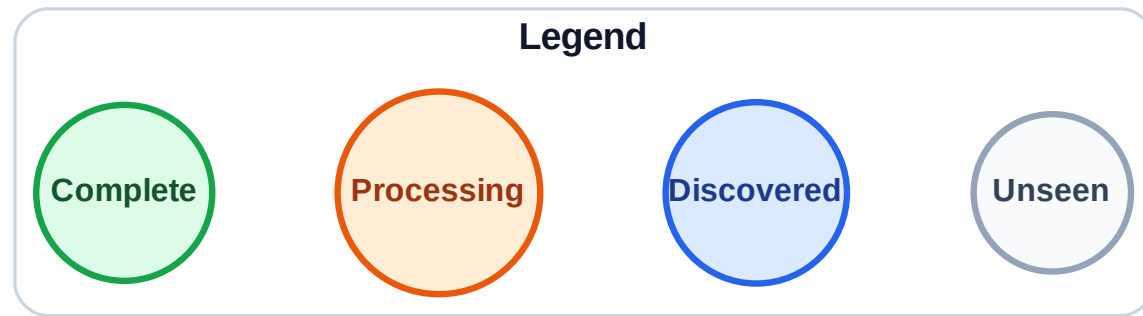
Queue: F → H → E → K → M → N

Visited: A, B, C

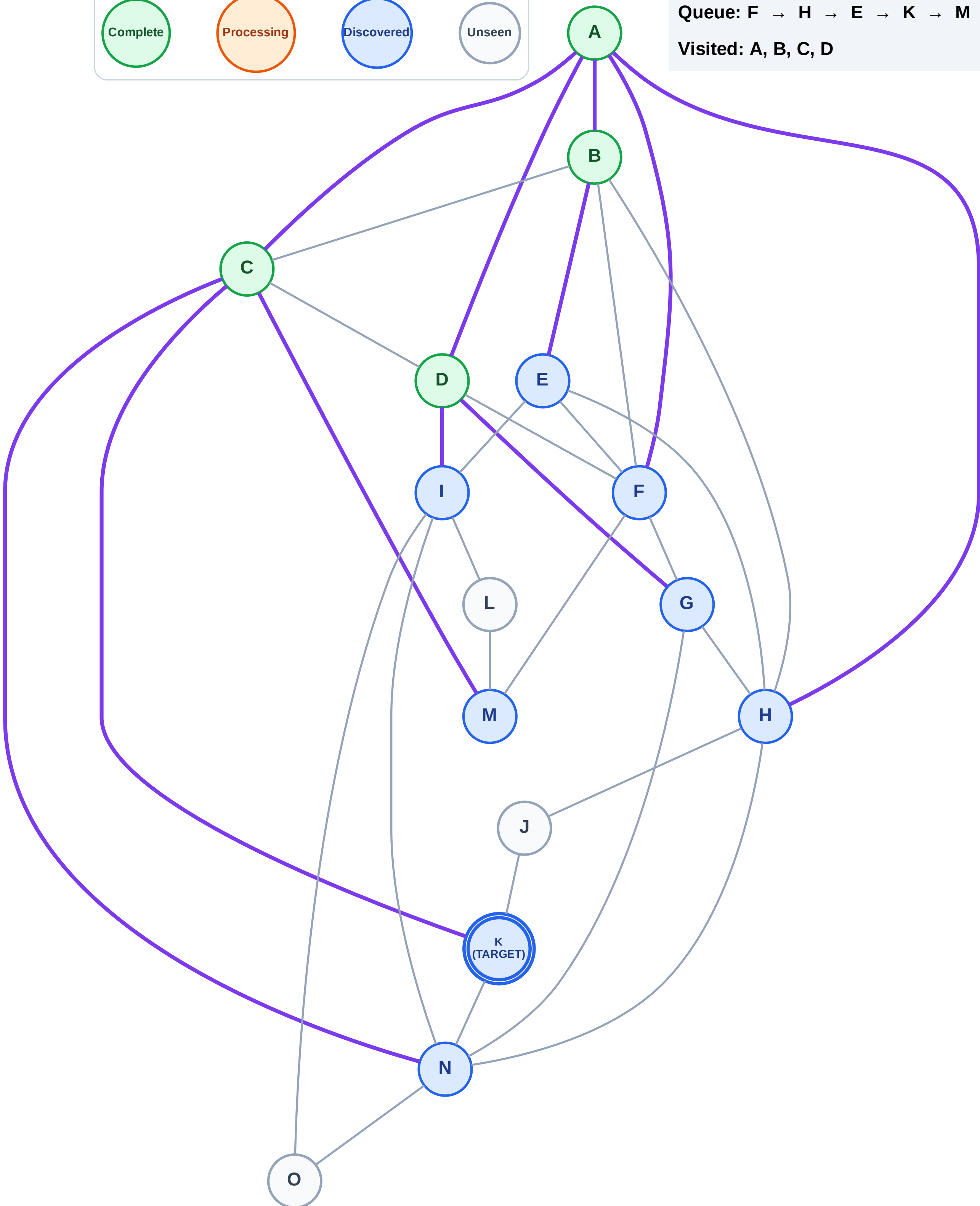


BFS Traversal

Step 9: Discover G, I from D



Queue: F → H → E → K → M → N → G → I
Visited: A, B, C, D



BFS Traversal

Step 10: Process node F

Legend

Complete

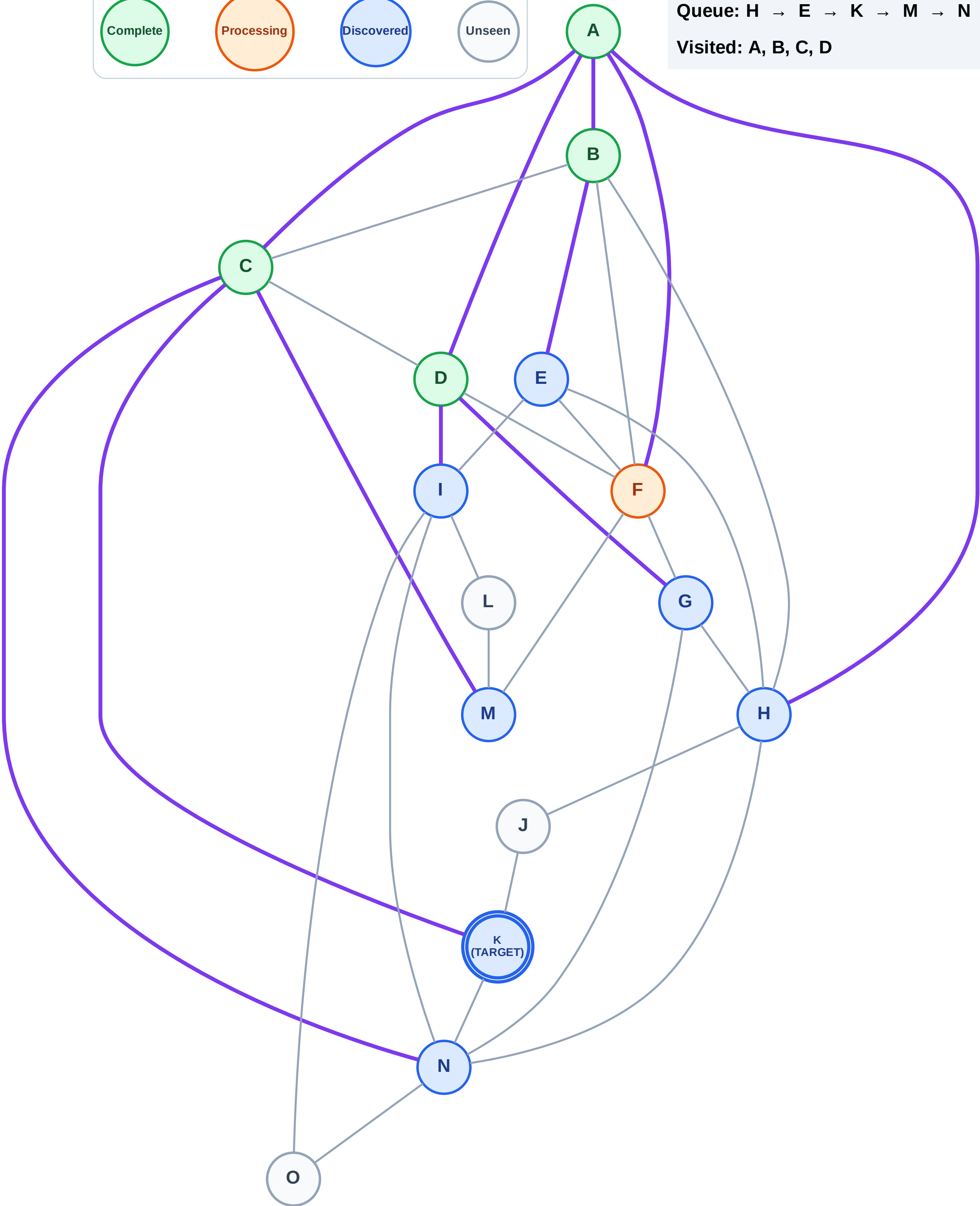
Processing

Discovered

Unseen

Queue: H → E → K → M → N → G → I

Visited: A, B, C, D



BFS Traversal

Step 11: No new neighbors from F

Legend

Complete

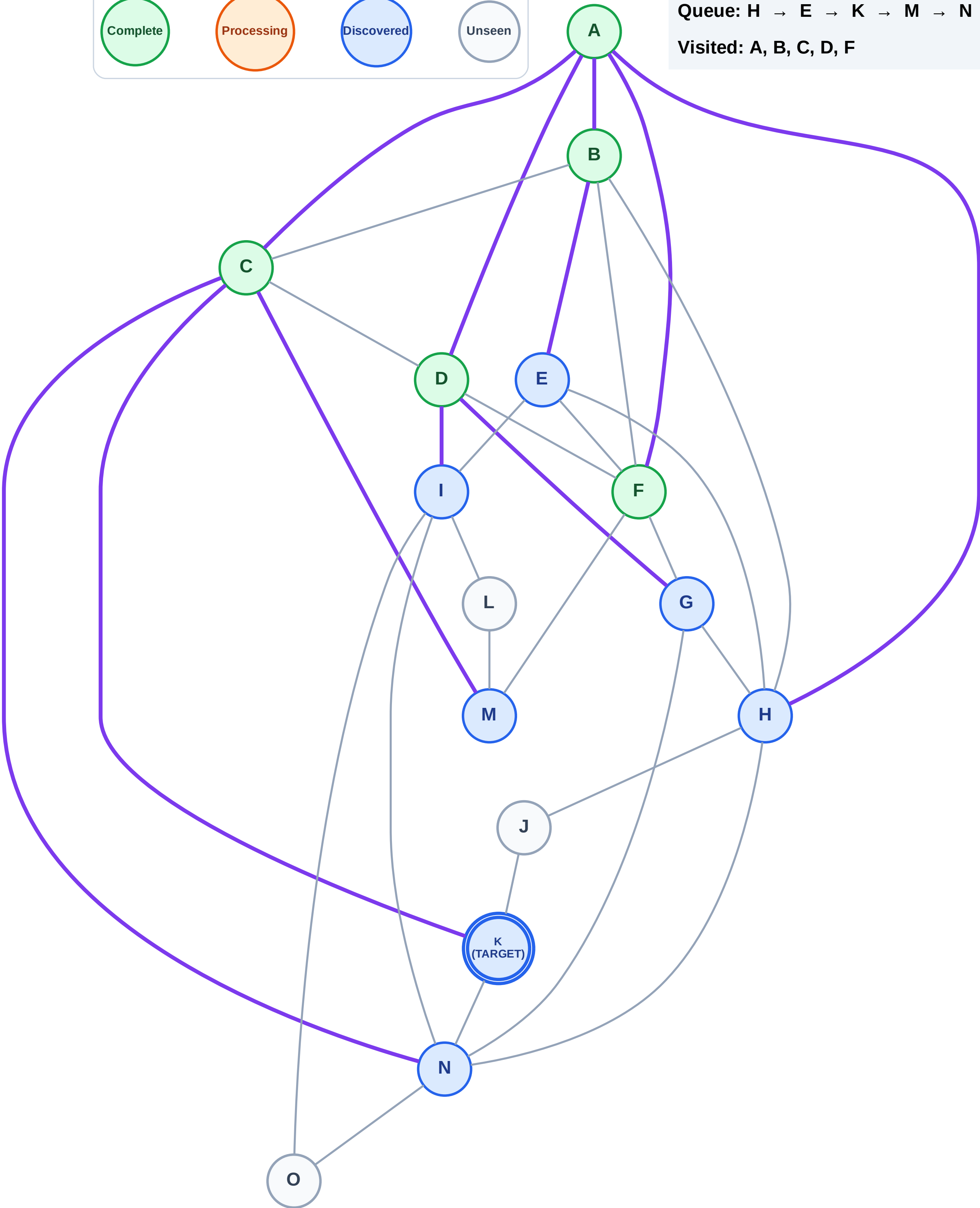
Processing

Discovered

Unseen

Queue: H → E → K → M → N → G → I

Visited: A, B, C, D, F



BFS Traversal

Step 12: Process node H

Legend

Complete

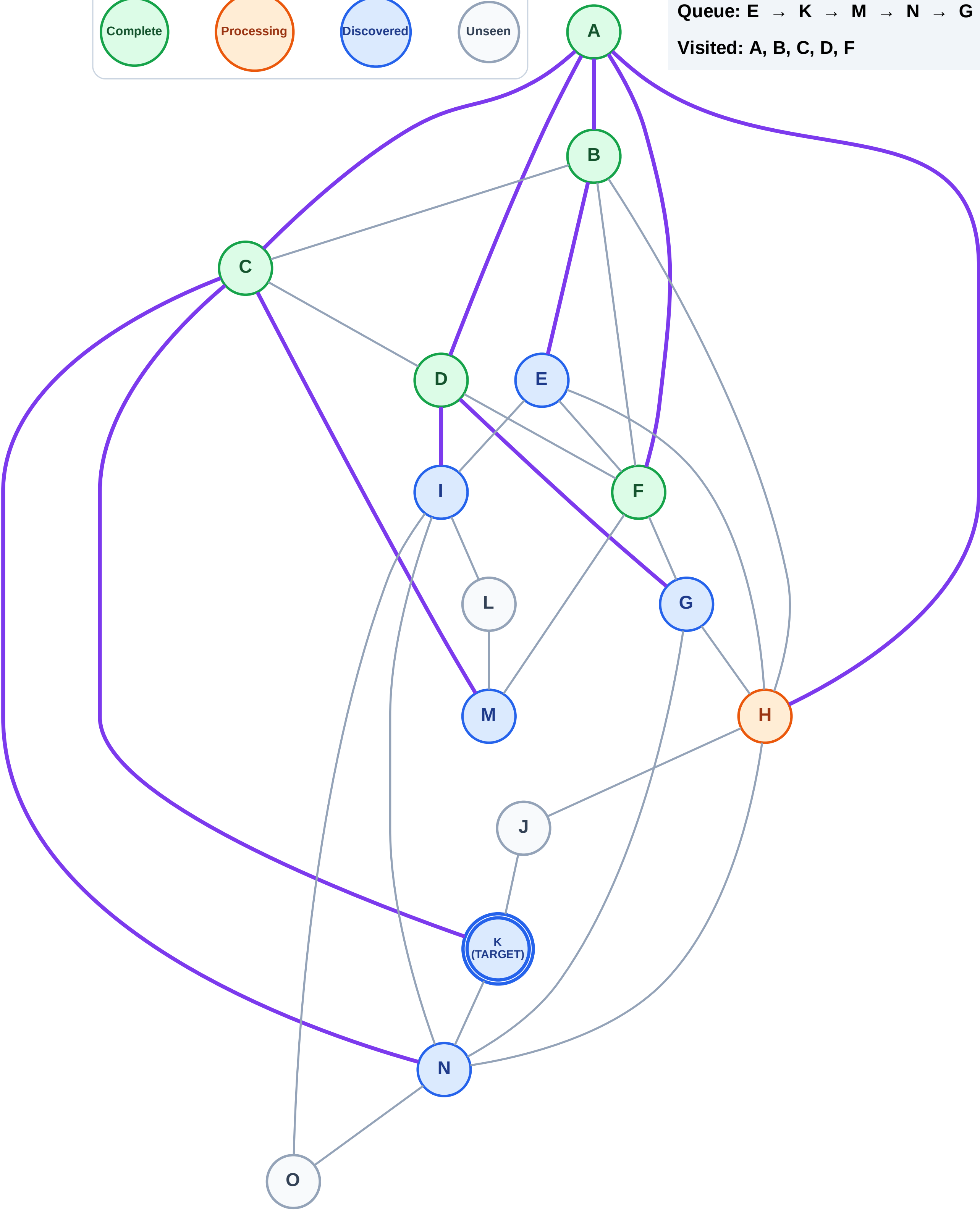
Processing

Discovered

Unseen

Queue: E → K → M → N → G → I

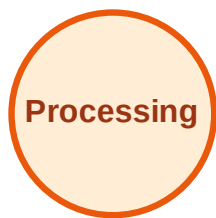
Visited: A, B, C, D, F



BFS Traversal

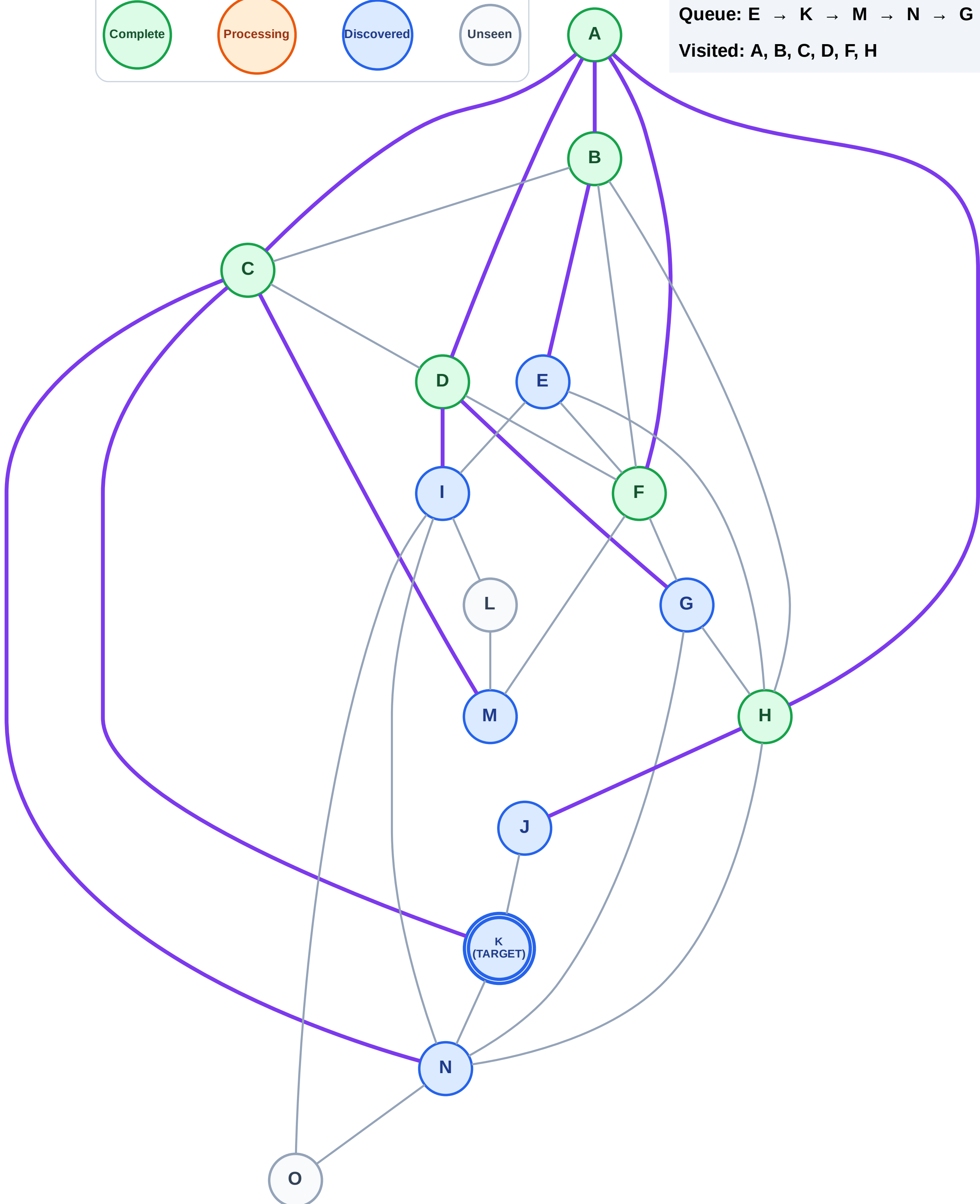
Step 13: Discover J from H

Legend



Queue: E → K → M → N → G → I → J

Visited: A, B, C, D, F, H



BFS Traversal

Step 14: Process node E

Legend

Complete

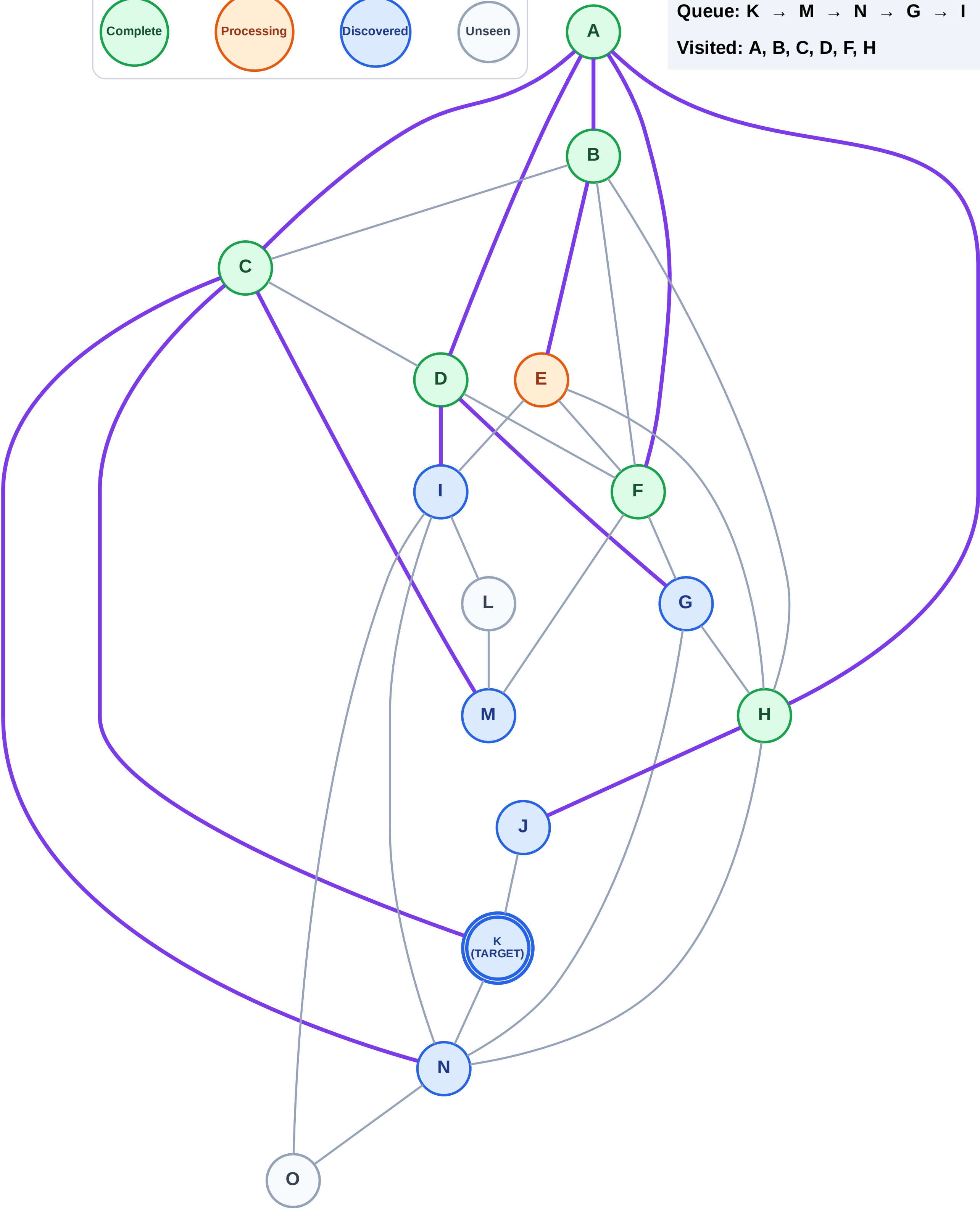
Processing

Discovered

Unseen

Queue: K → M → N → G → I → J

Visited: A, B, C, D, F, H



BFS Traversal

Step 15: No new neighbors from E

Legend

Complete

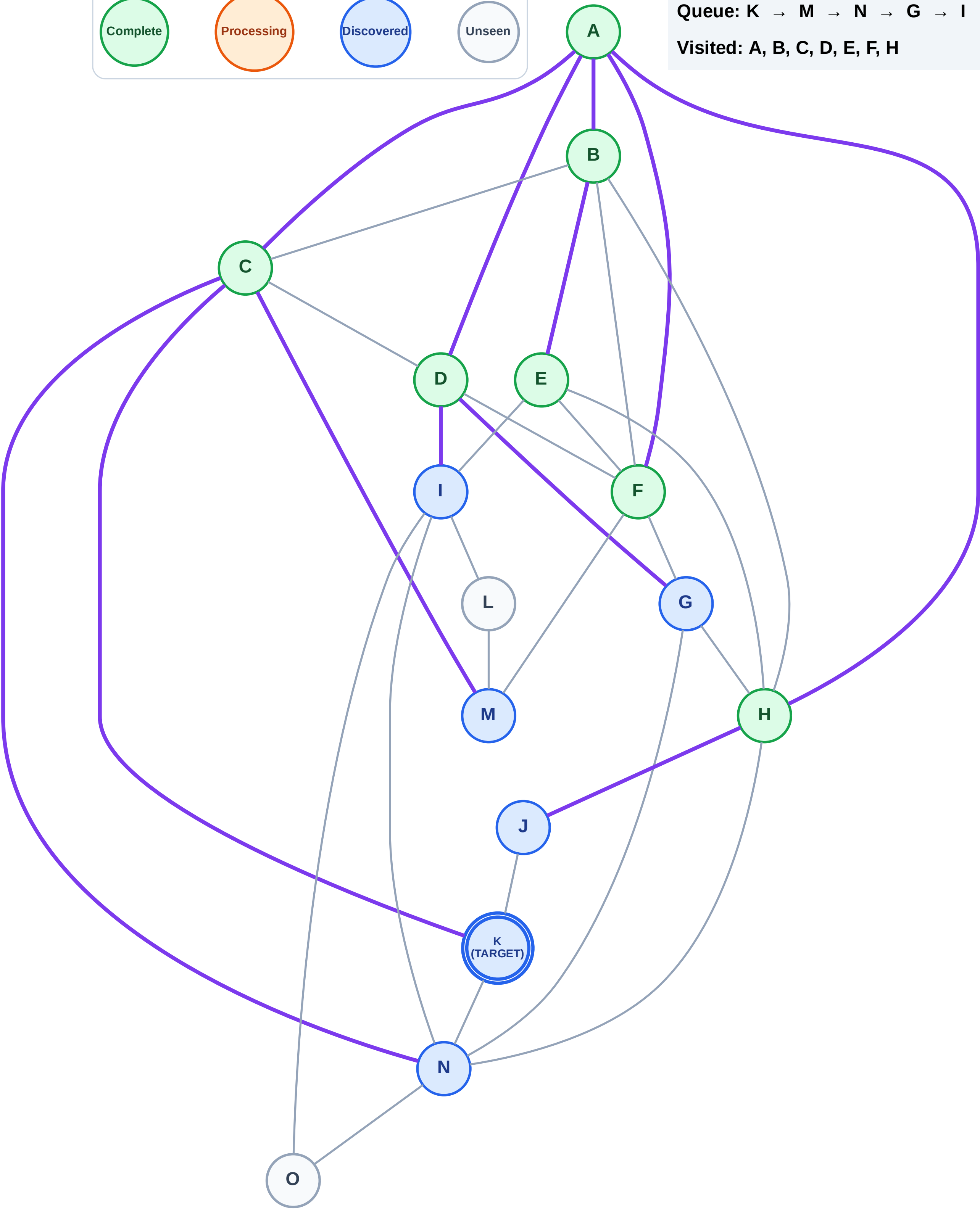
Processing

Discovered

Unseen

Queue: K → M → N → G → I → J

Visited: A, B, C, D, E, F, H



BFS Traversal

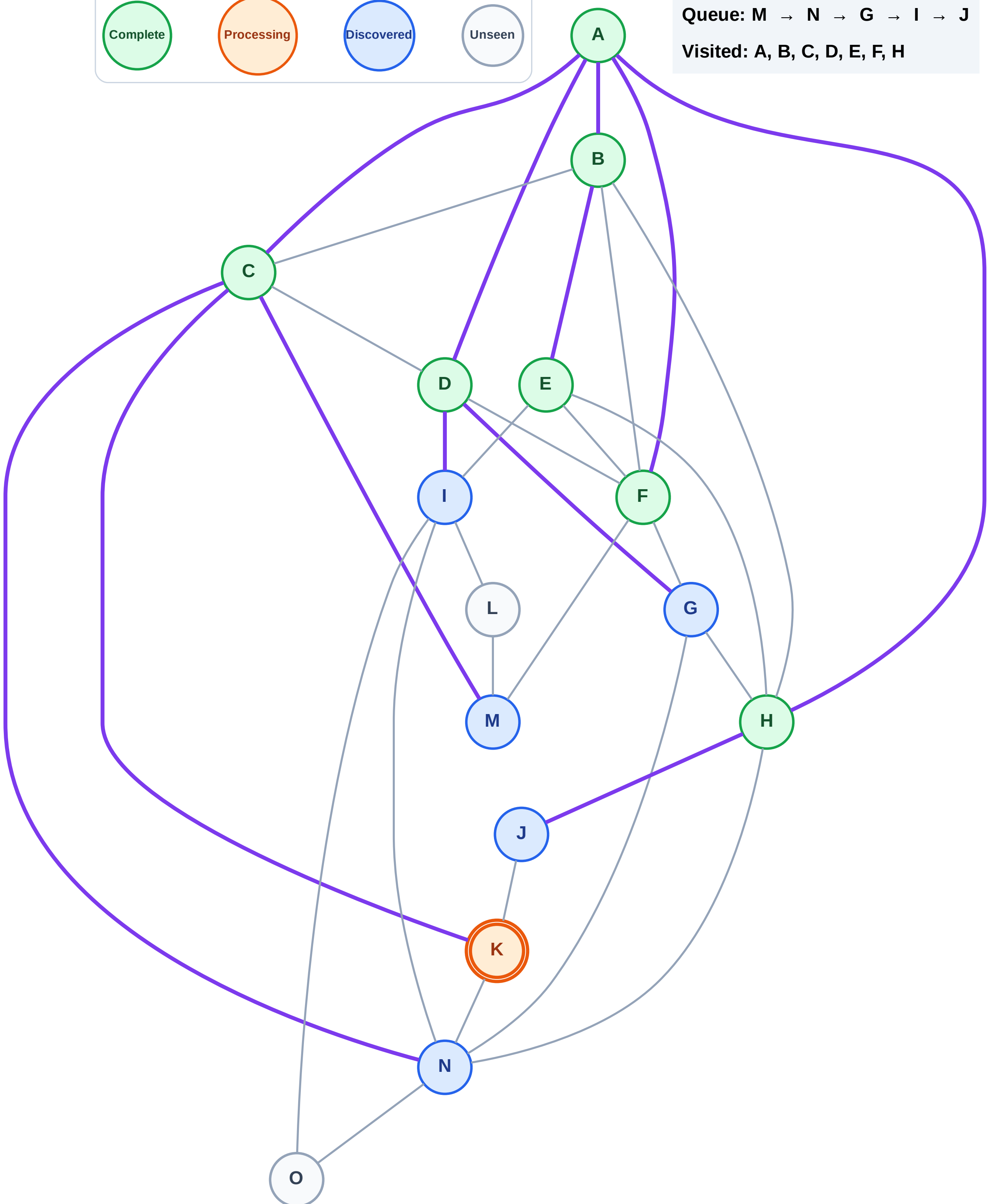
Step 16: Process node K

Legend



Queue: M → N → G → I → J

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

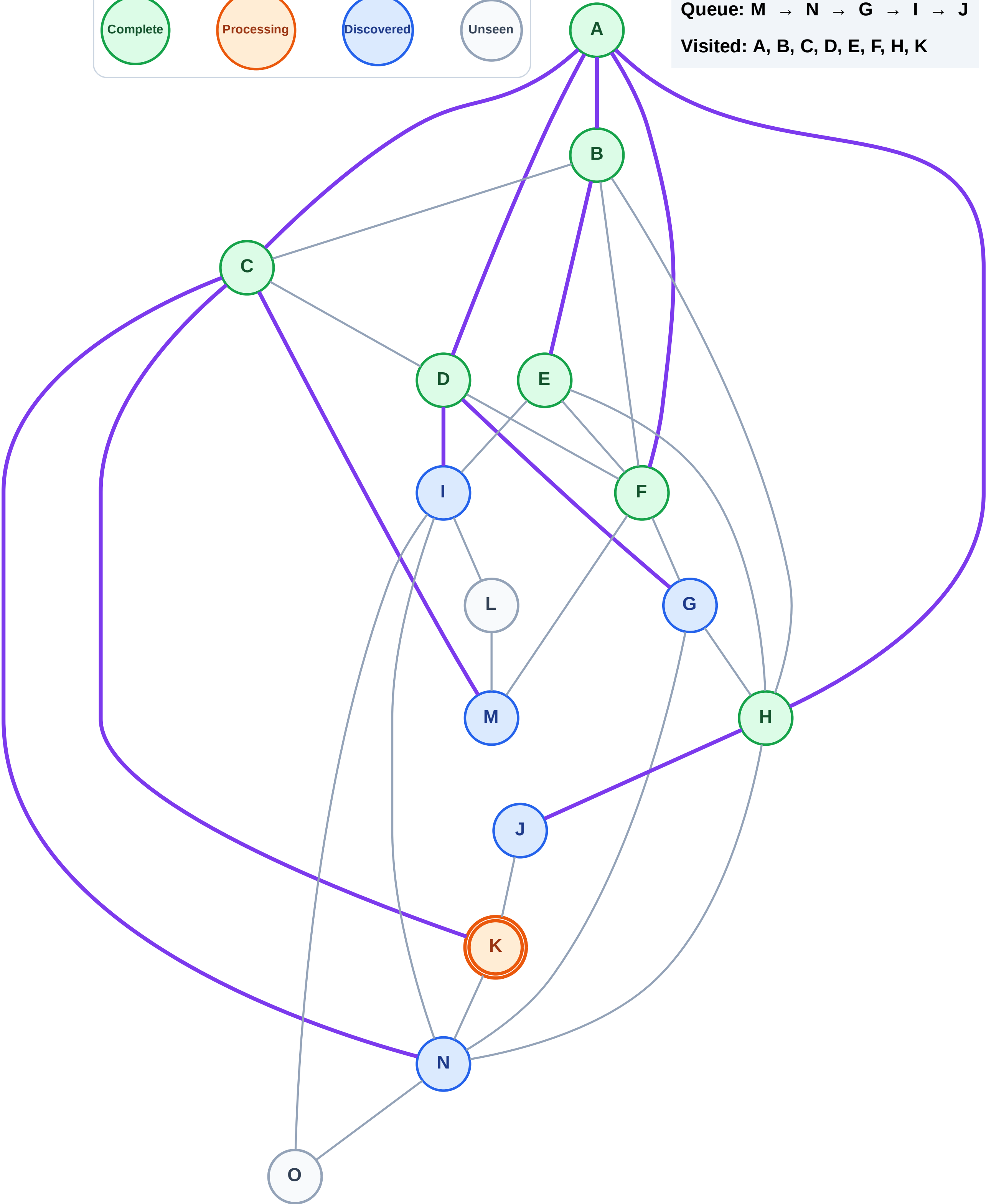
Processing

Discovered

Unseen

Queue: M → N → G → I → J

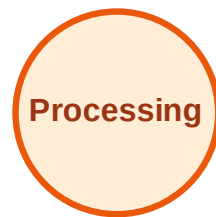
Visited: A, B, C, D, E, F, H, K



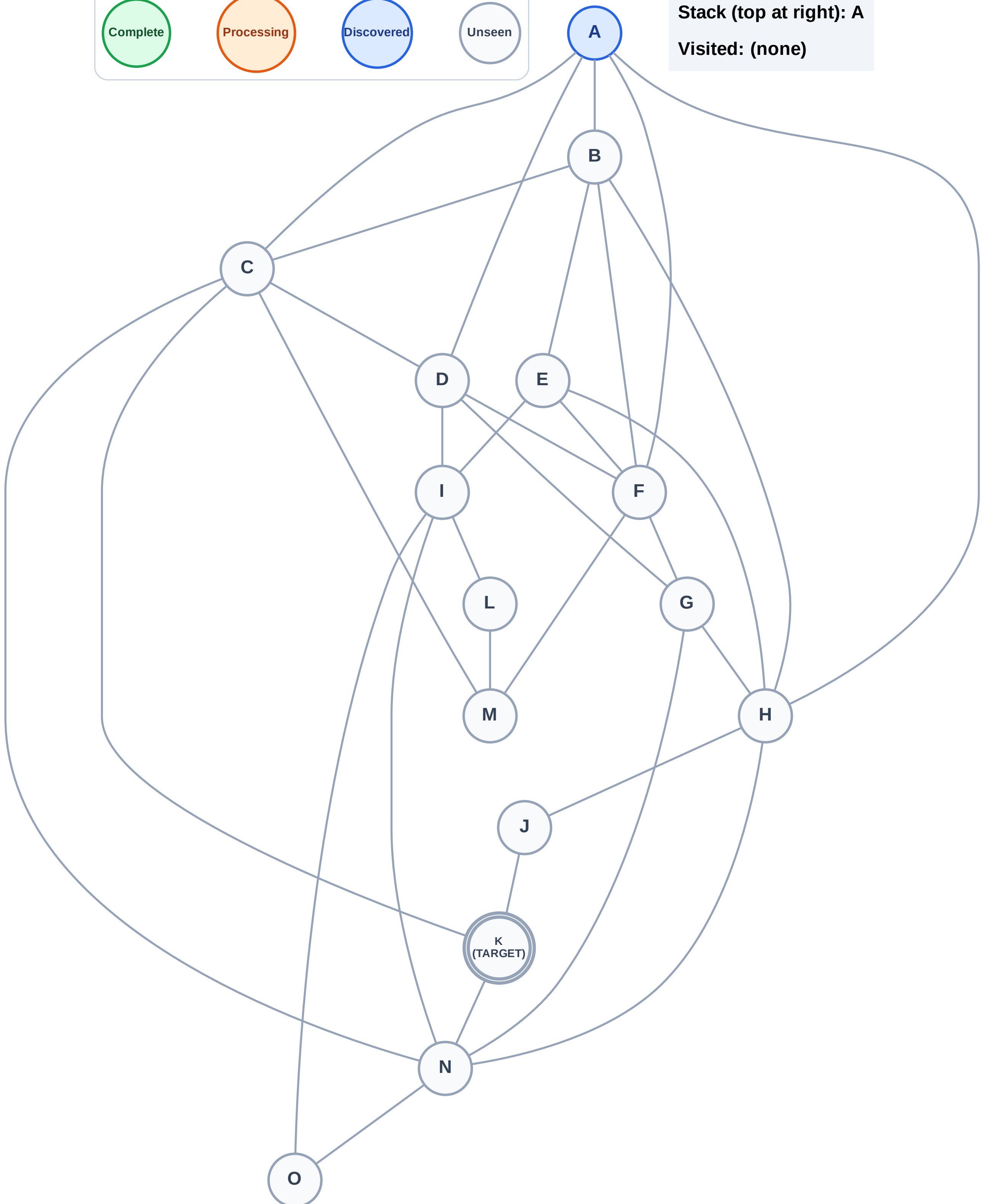
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

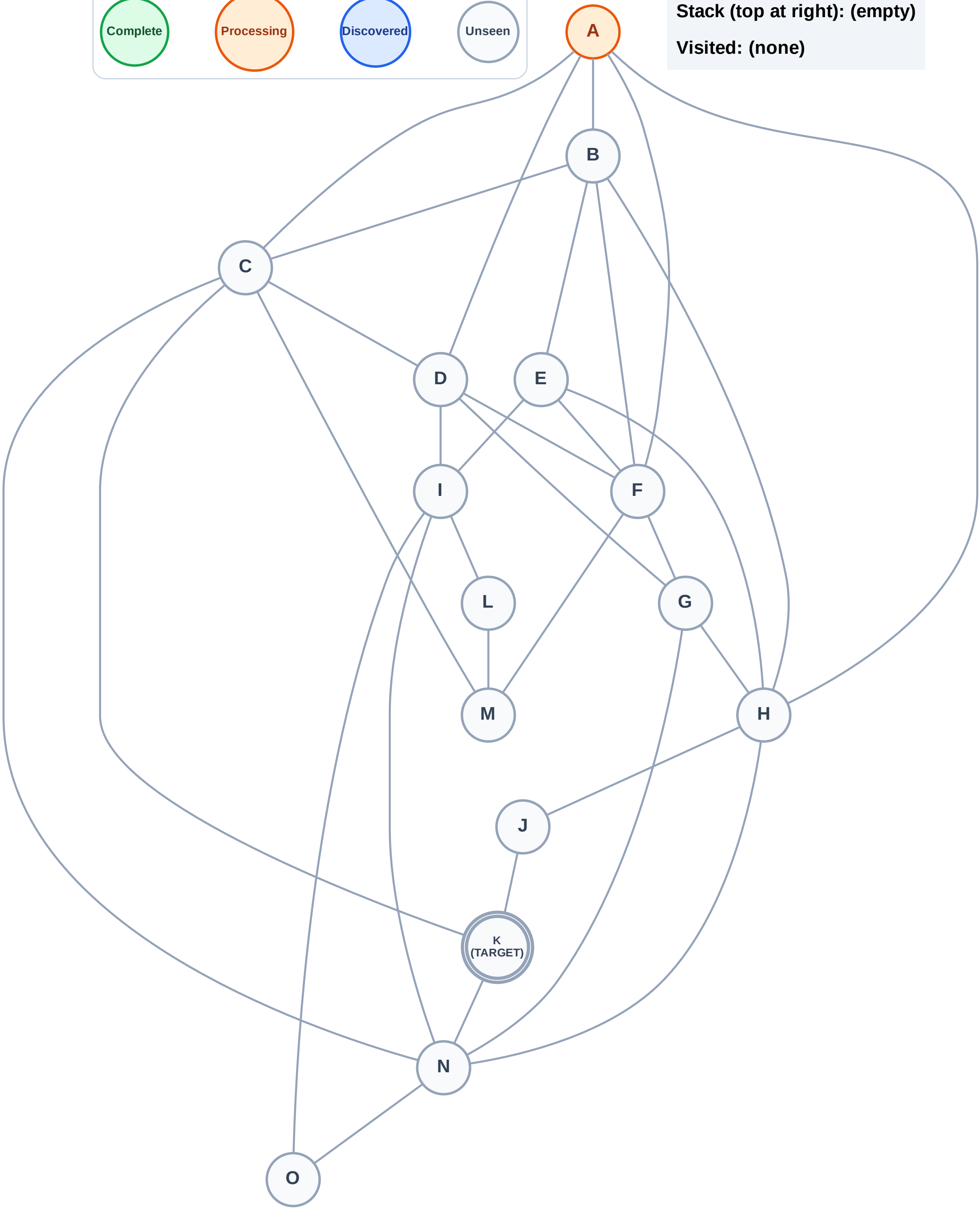
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, F, D, C, B from A

Legend

Complete

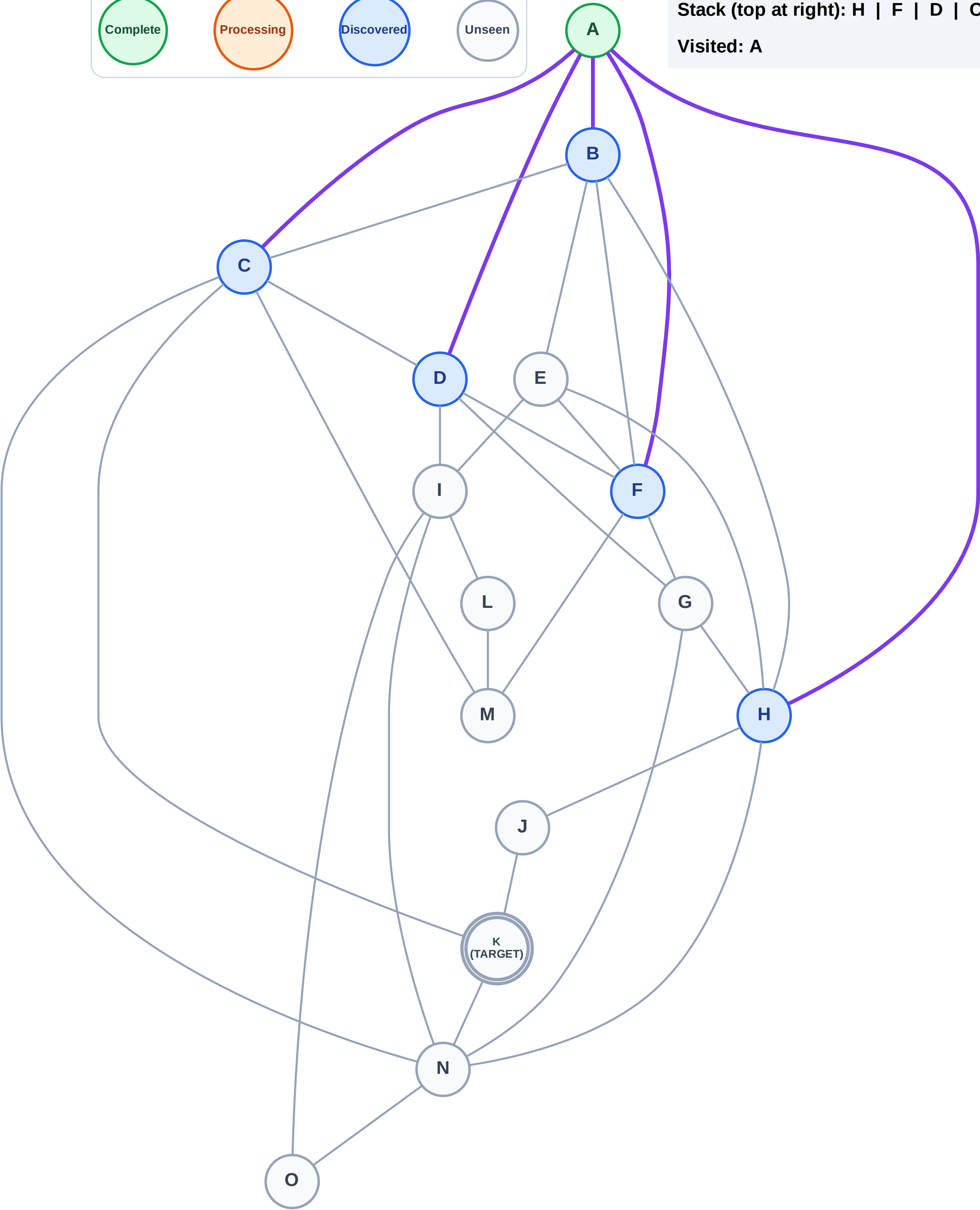
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

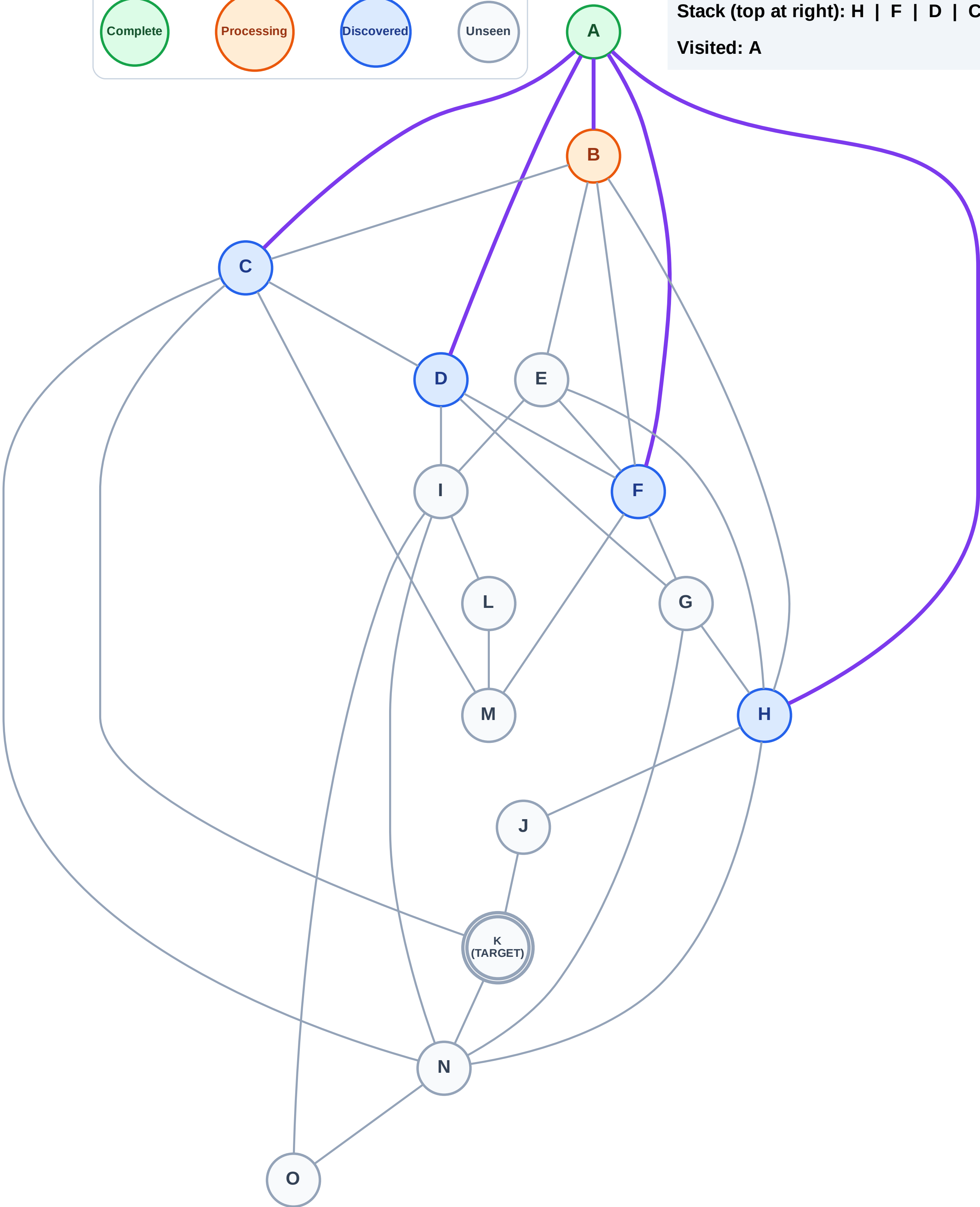
Processing

Discovered

Unseen

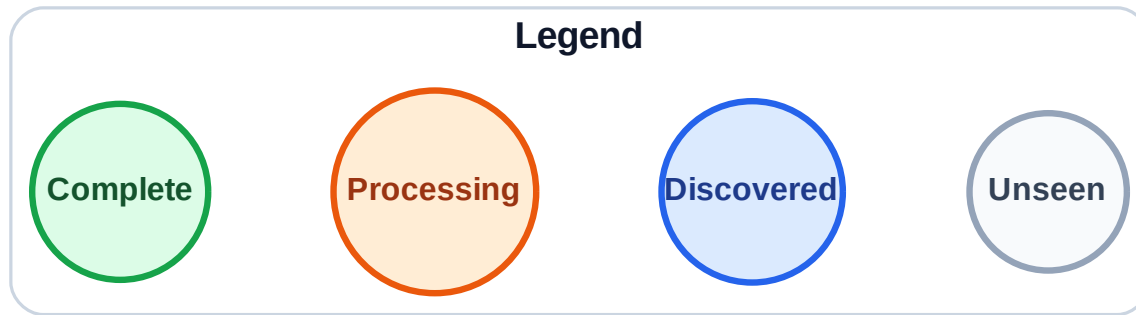
Stack (top at right): H | F | D | C

Visited: A

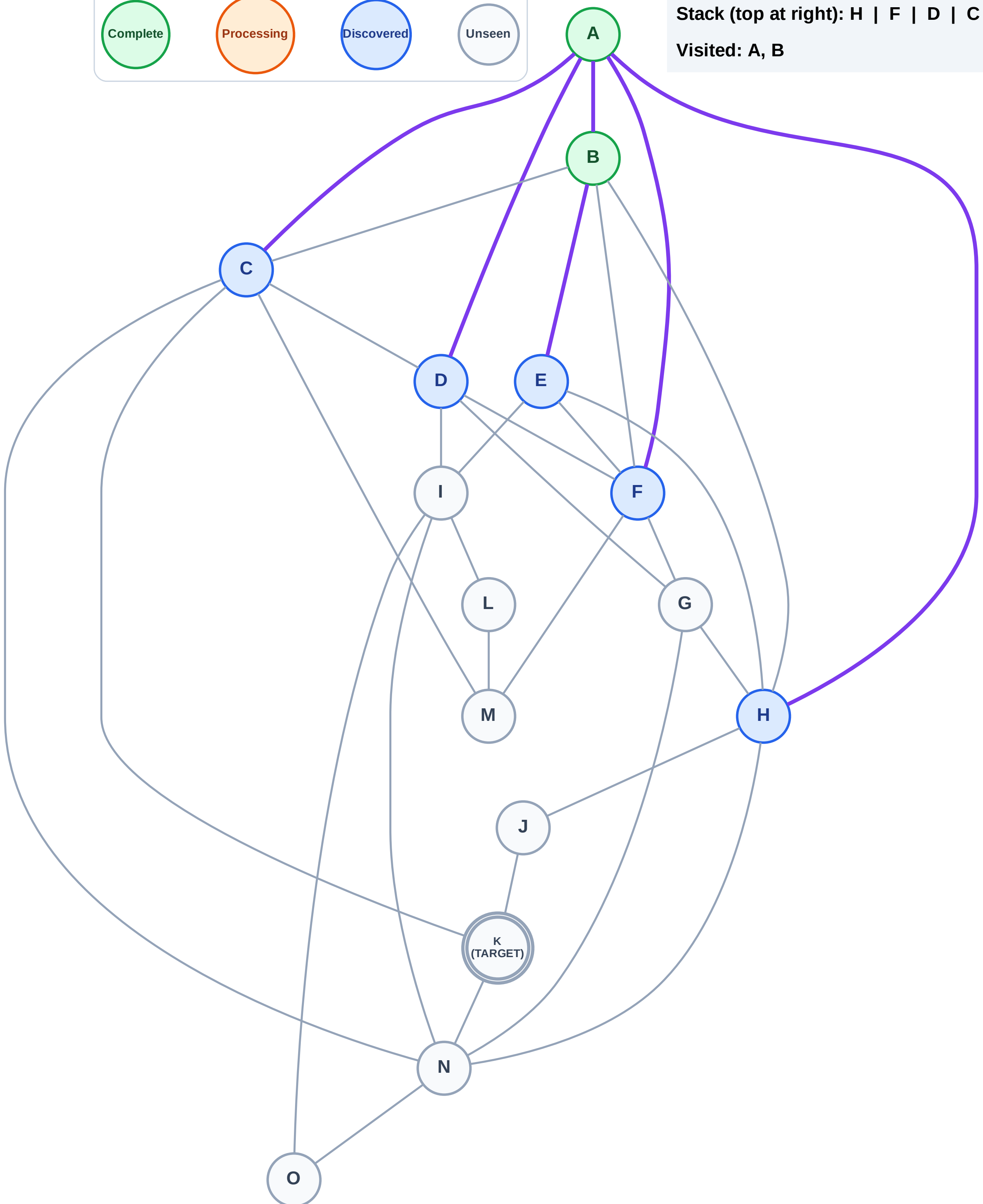


DFS Traversal

Step 5: Discover E from B



Stack (top at right): H | F | D | C | E
Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

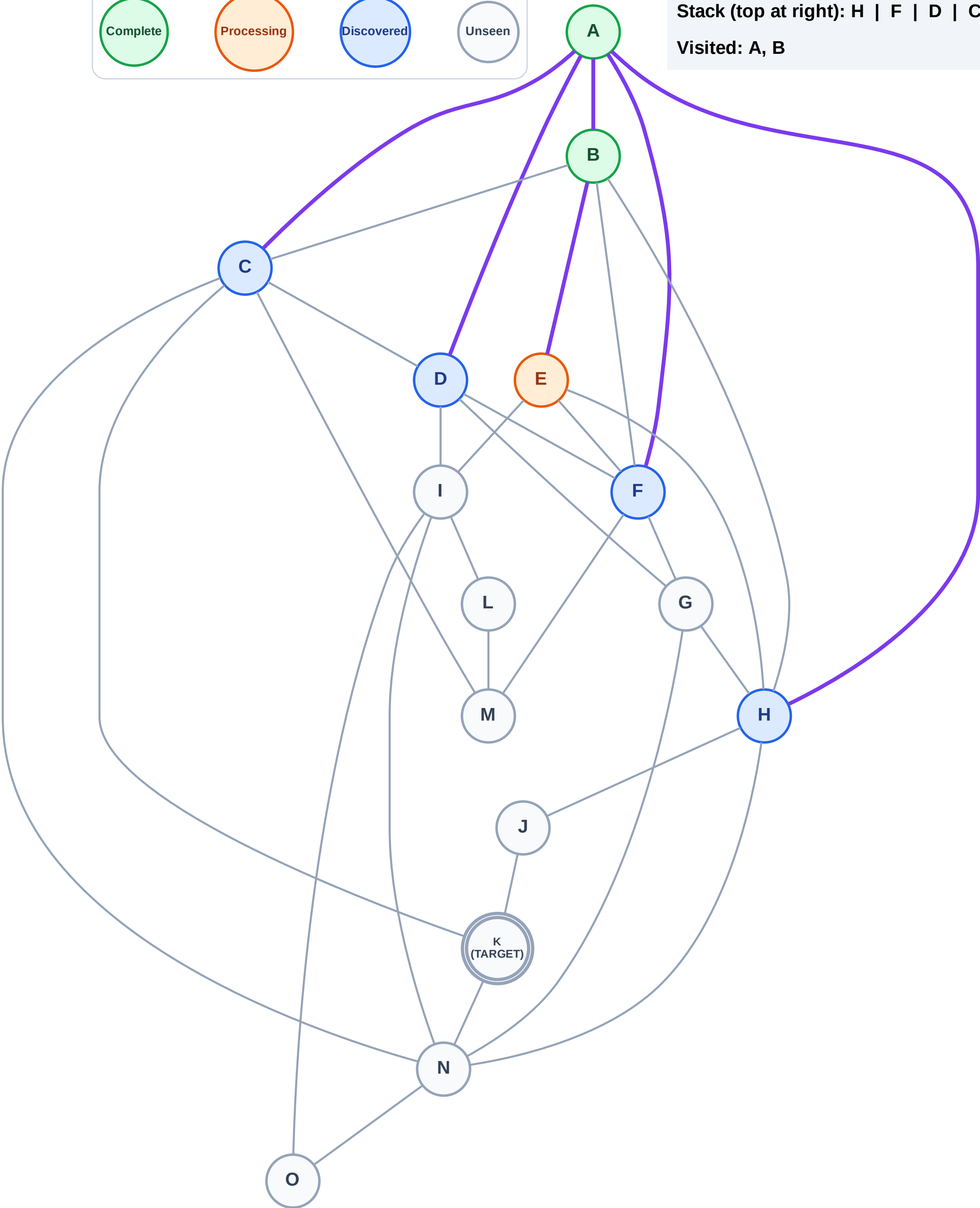
Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C
Visited: A, B



DFS Traversal

Step 7: Discover I from E

Legend

Complete

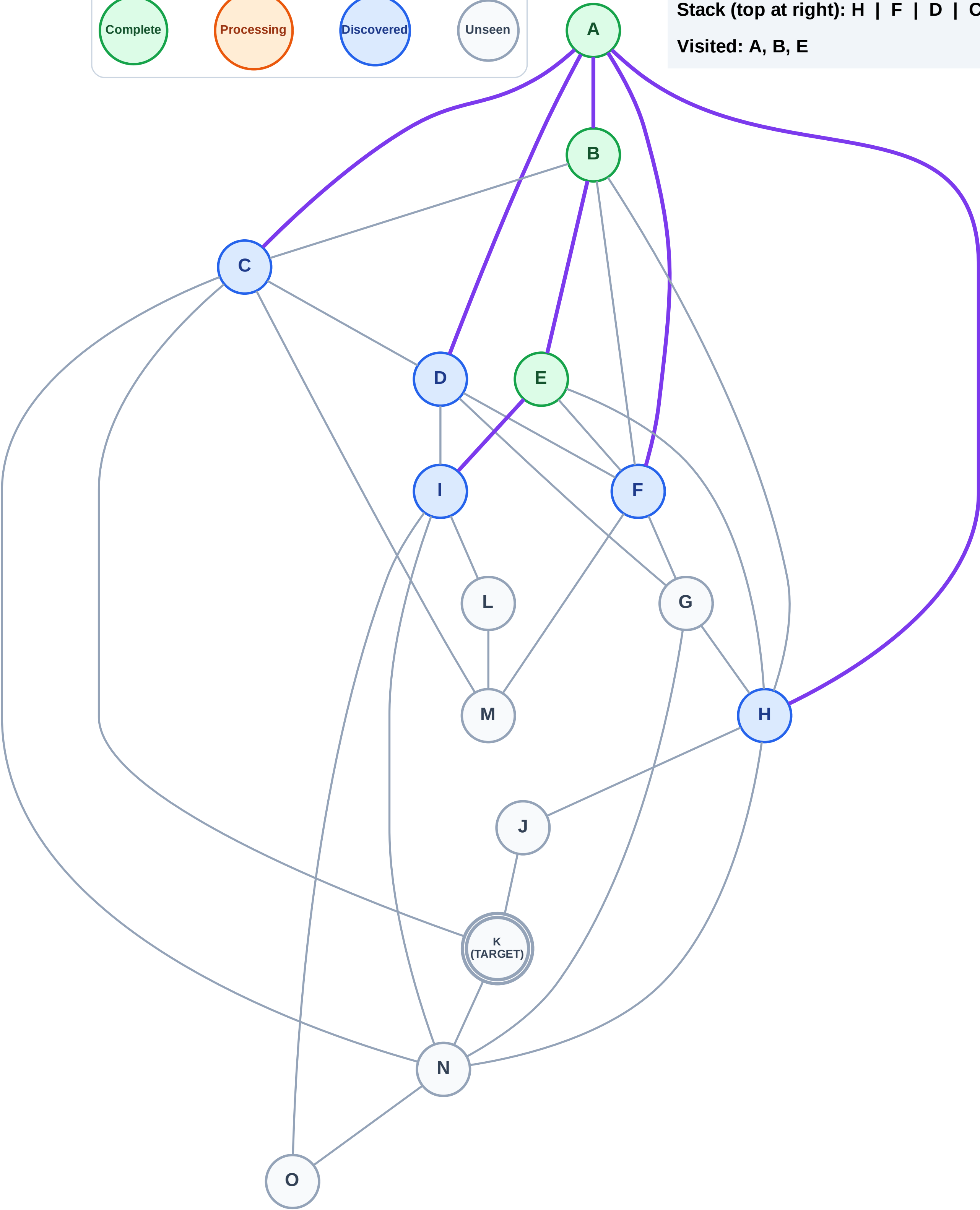
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | I

Visited: A, B, E



DFS Traversal

Step 8: Process node I

Legend

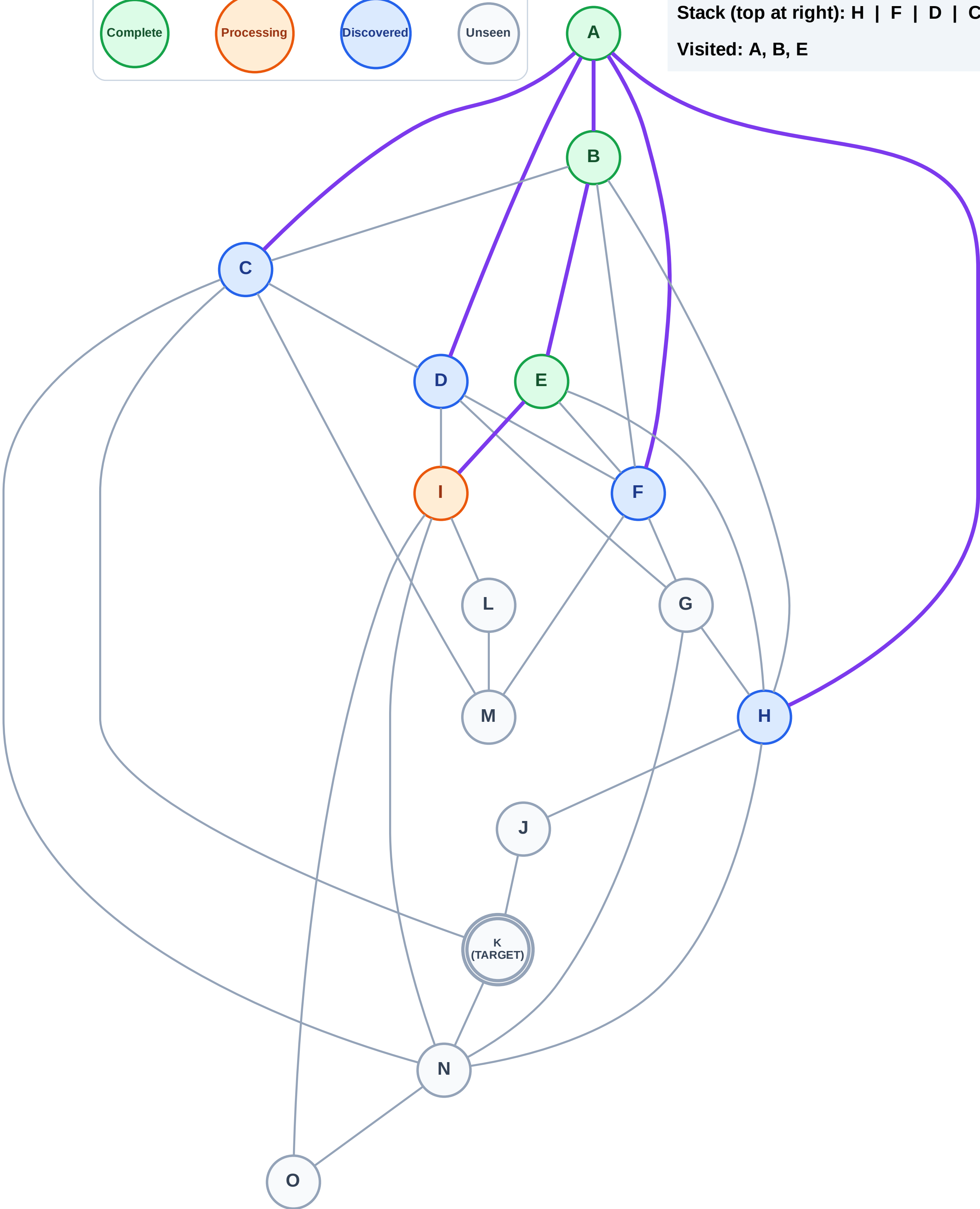
Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C
Visited: A, B, E



DFS Traversal

Step 9: Discover O, N, L from I

Legend

Complete

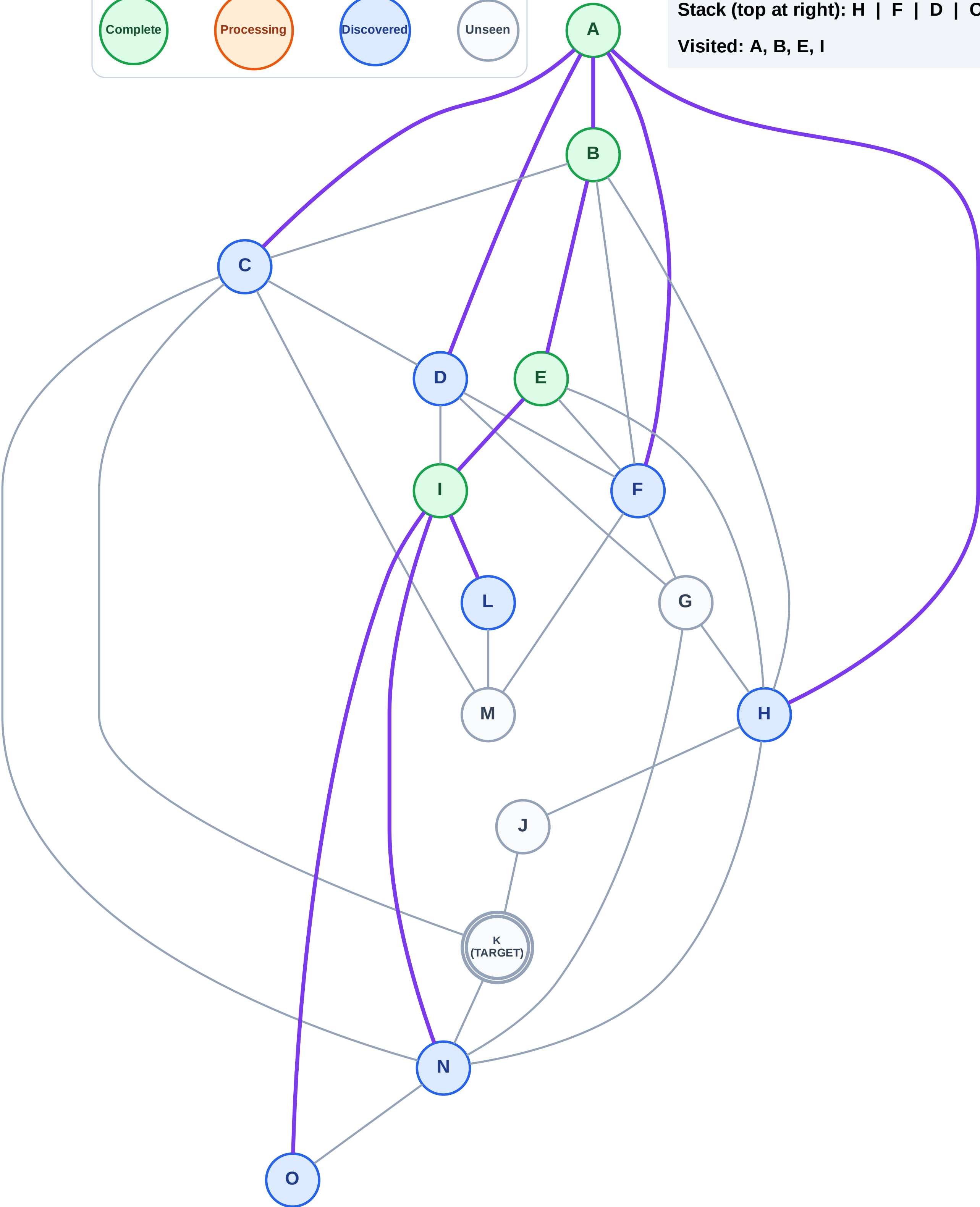
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N | L

Visited: A, B, E, I



DFS Traversal

Step 10: Process node L

Legend

Complete

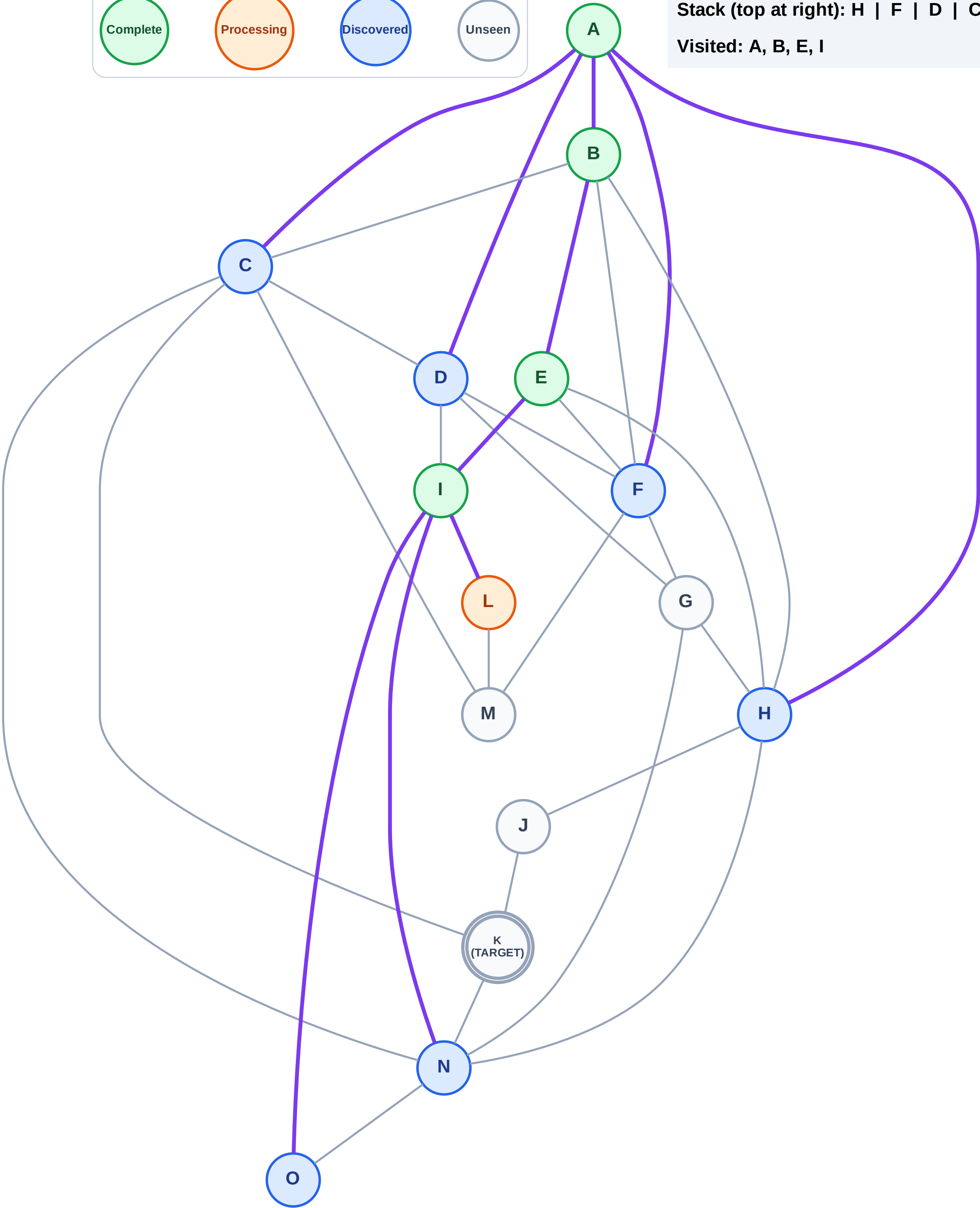
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N

Visited: A, B, E, I



DFS Traversal

Step 11: Discover M from L

Legend

Complete

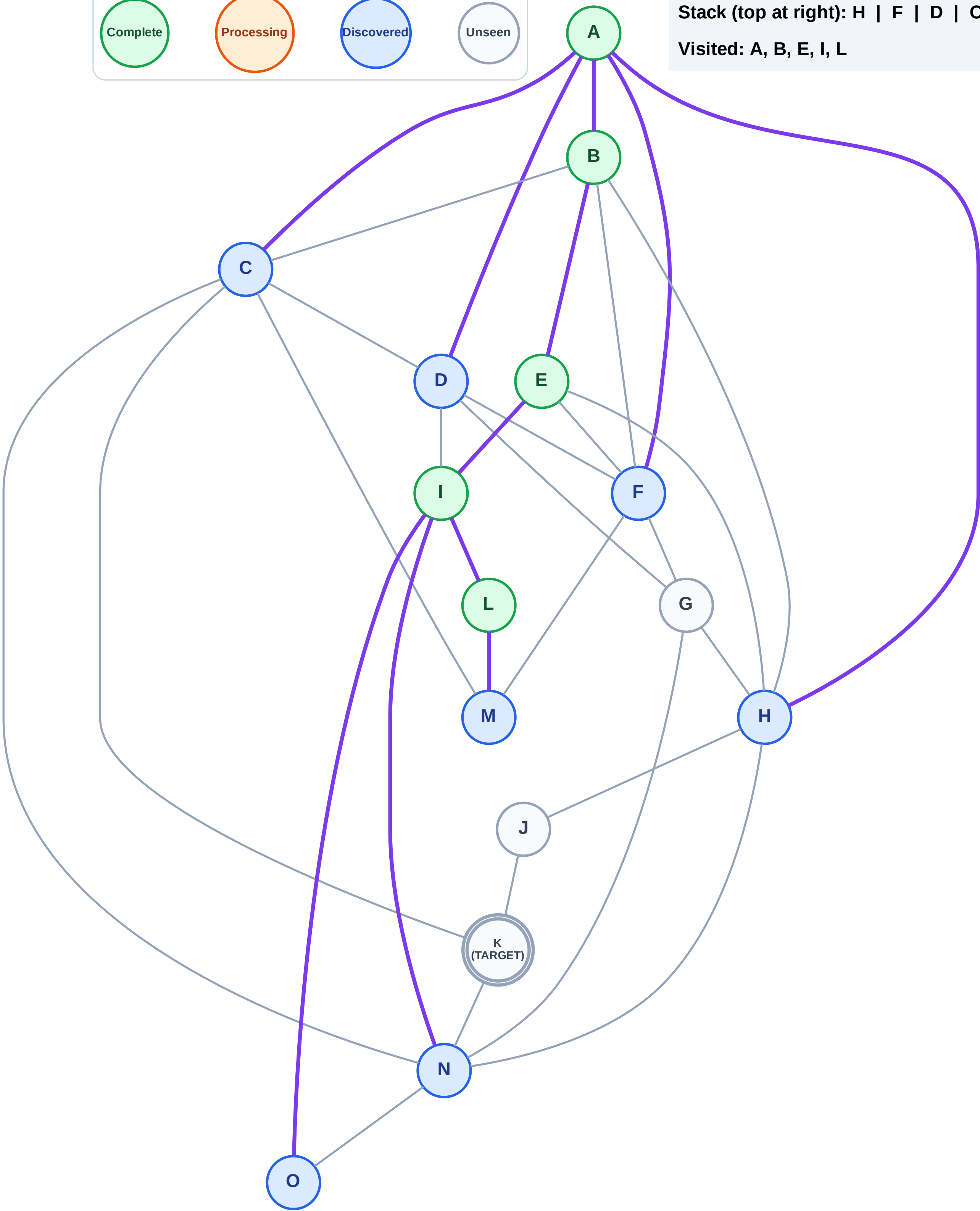
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N | M

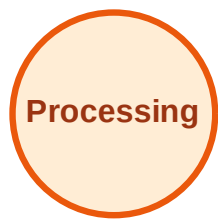
Visited: A, B, E, I, L



DFS Traversal

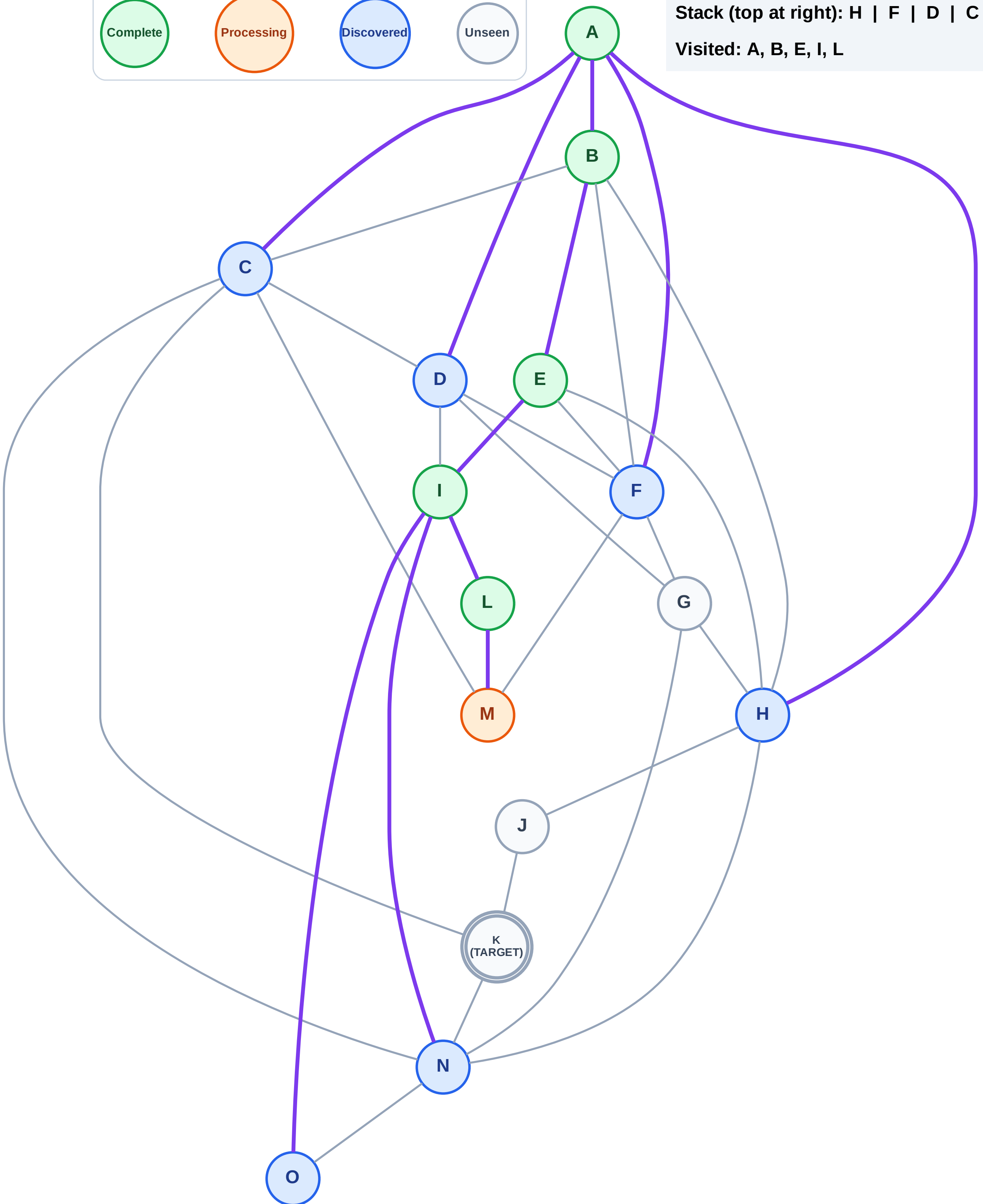
Step 12: Process node M

Legend



Stack (top at right): H | F | D | C | O | N

Visited: A, B, E, I, L



DFS Traversal

Step 13: No new neighbors from M

Legend

Complete

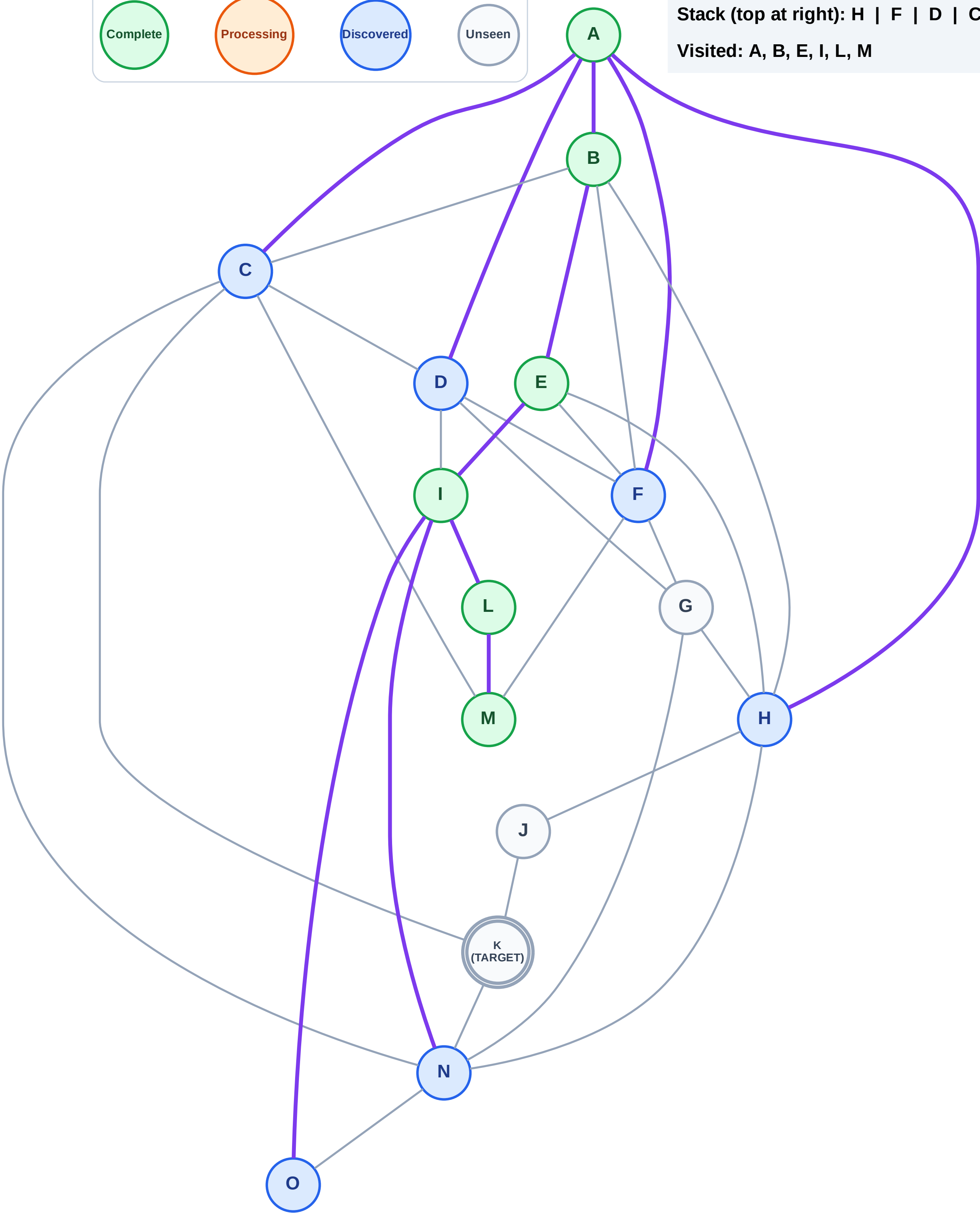
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N

Visited: A, B, E, I, L, M



DFS Traversal

Step 14: Process node N

Legend

Complete

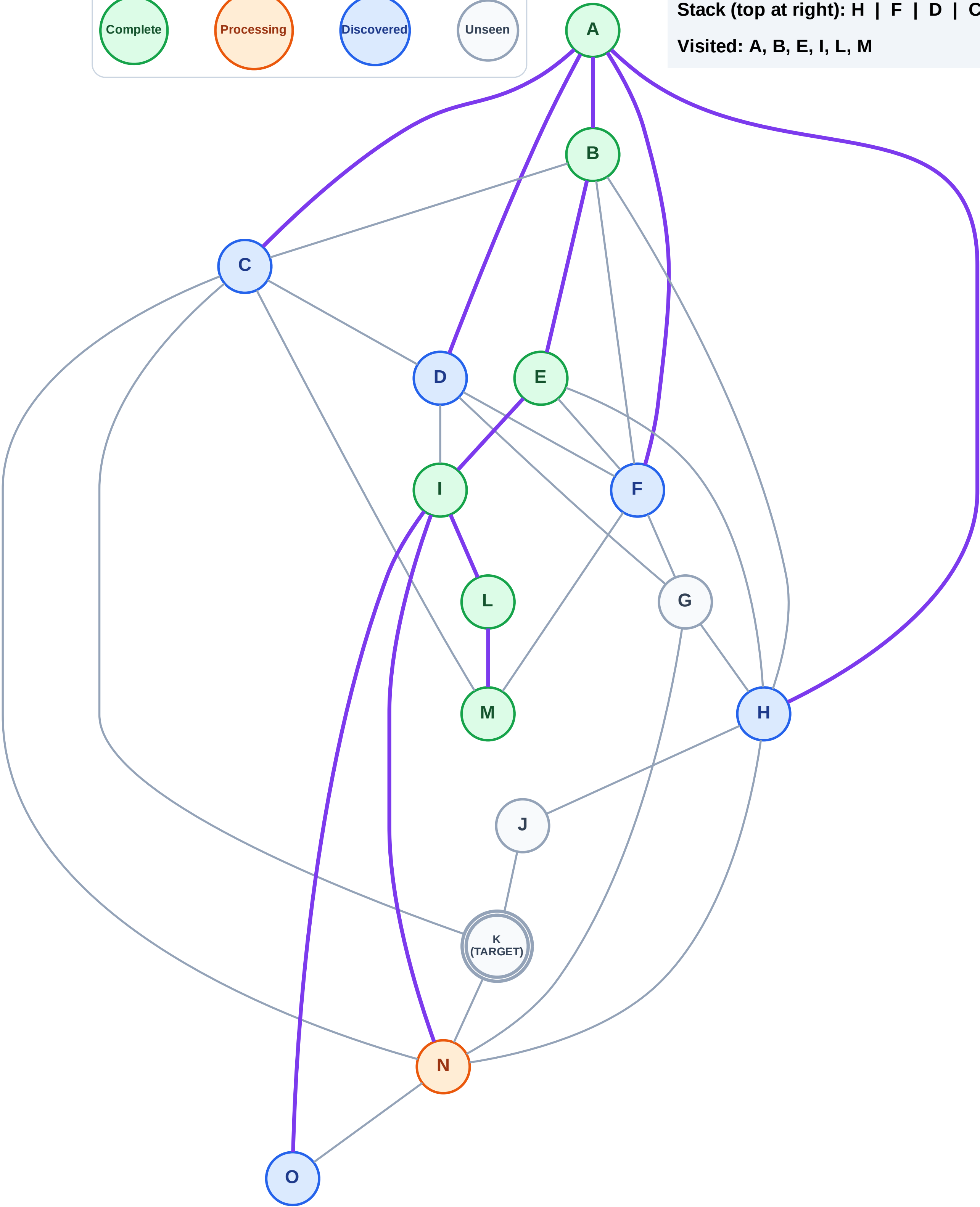
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O

Visited: A, B, E, I, L, M



DFS Traversal

Step 15: Discover K, G from N

Legend

Complete

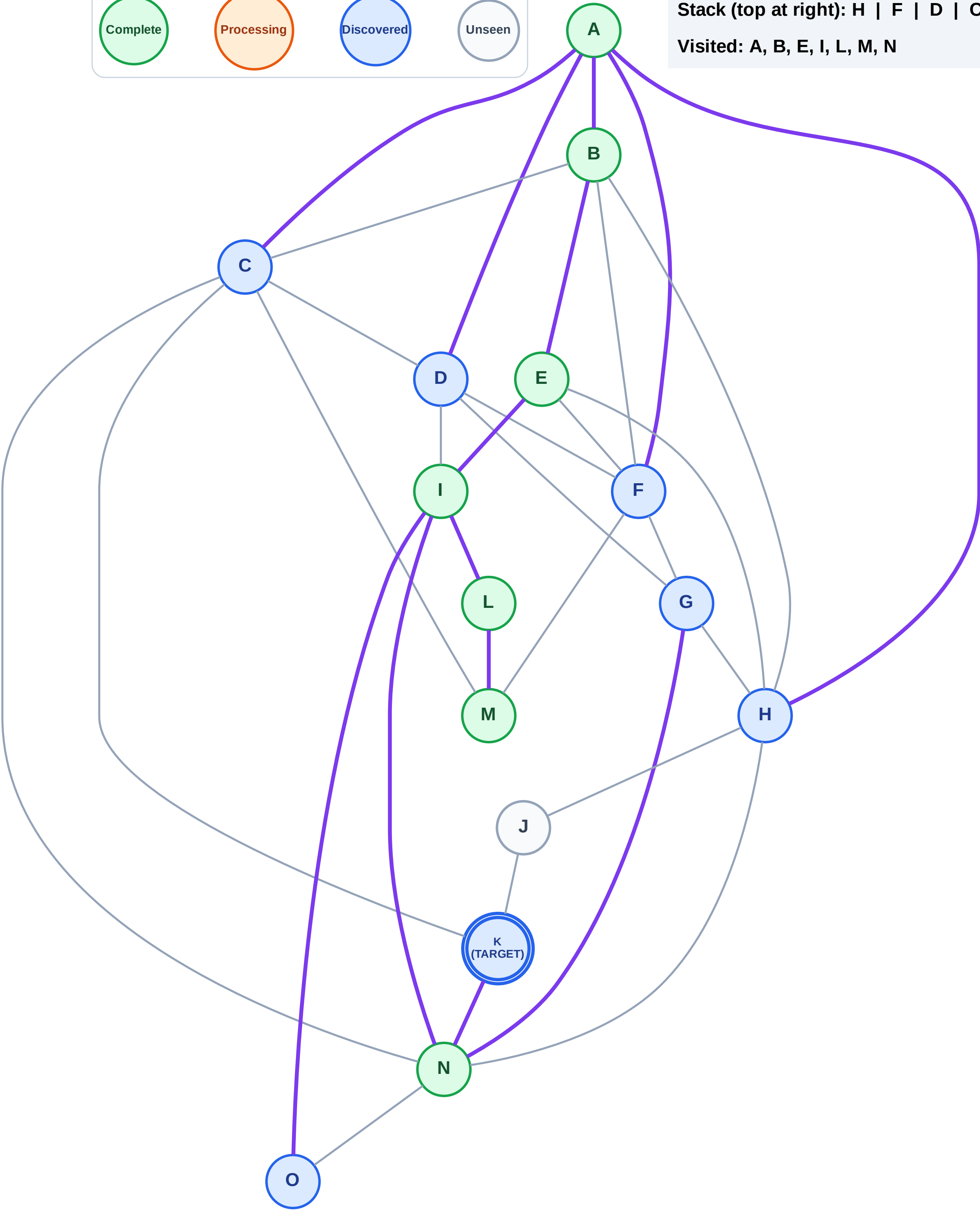
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | K | G

Visited: A, B, E, I, L, M, N



DFS Traversal

Step 16: Process node G

Legend

Complete

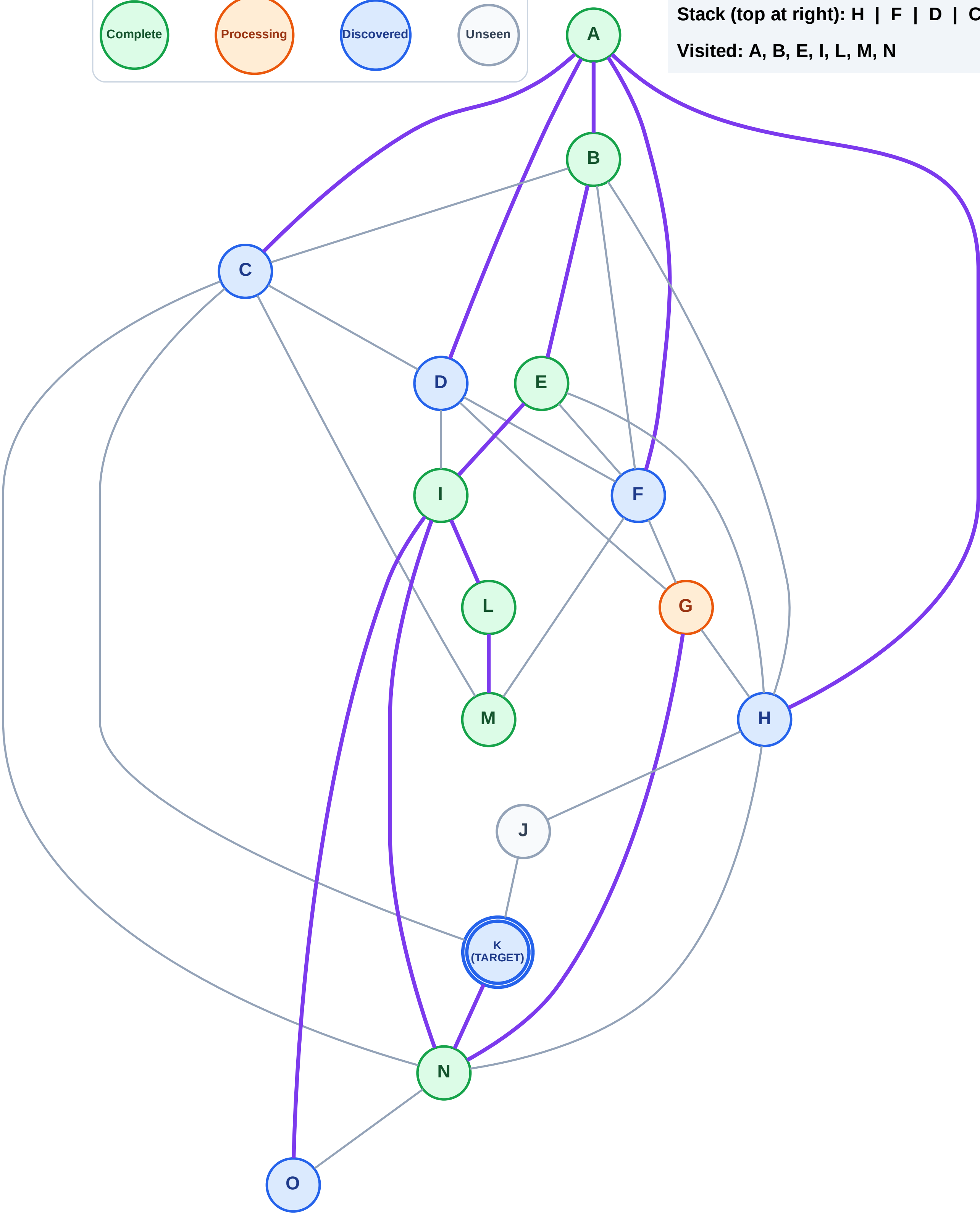
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | K

Visited: A, B, E, I, L, M, N



DFS Traversal

Step 17: No new neighbors from G

Legend

Complete

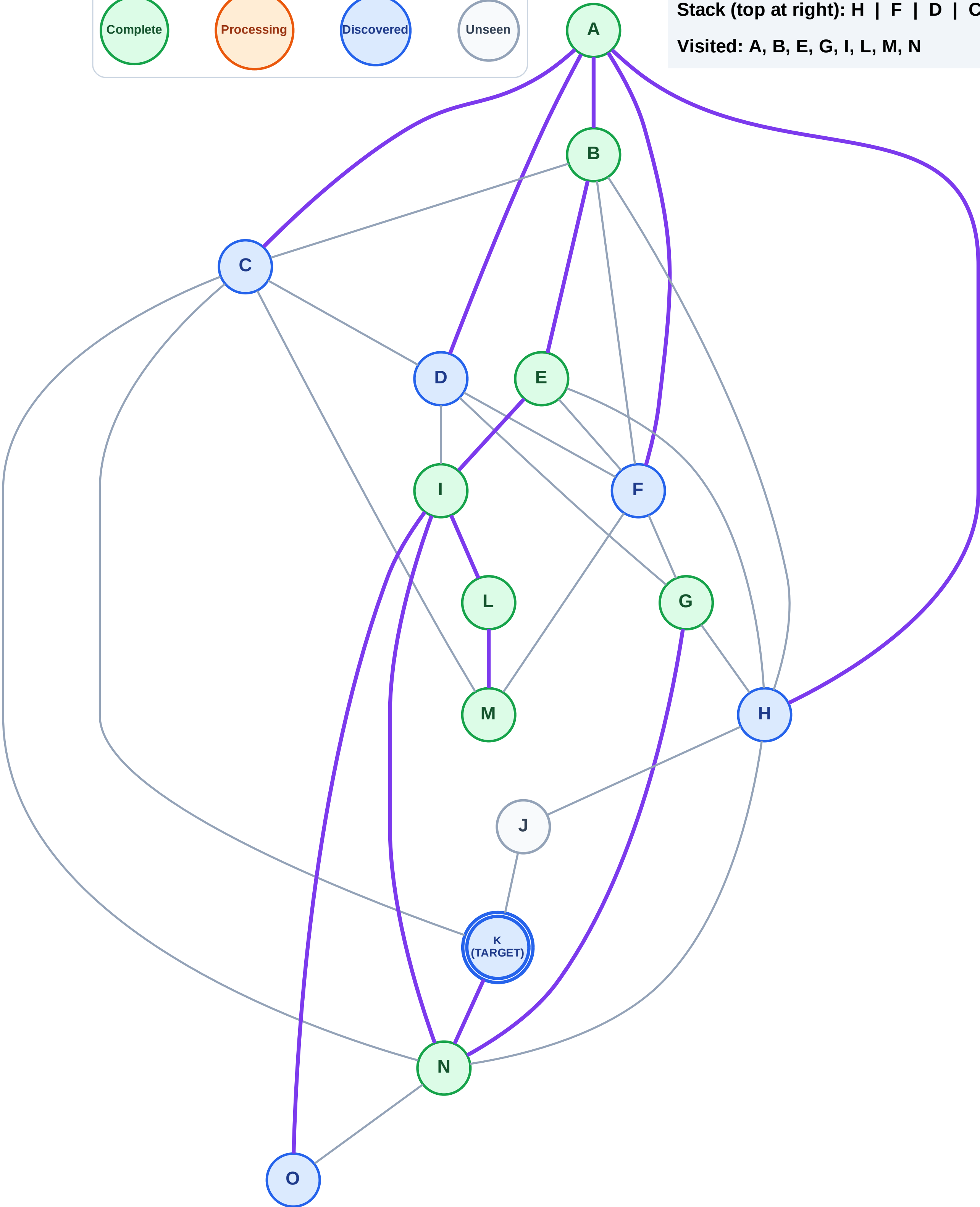
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | K

Visited: A, B, E, G, I, L, M, N



DFS Traversal

Step 18: Process node K

Legend

Complete

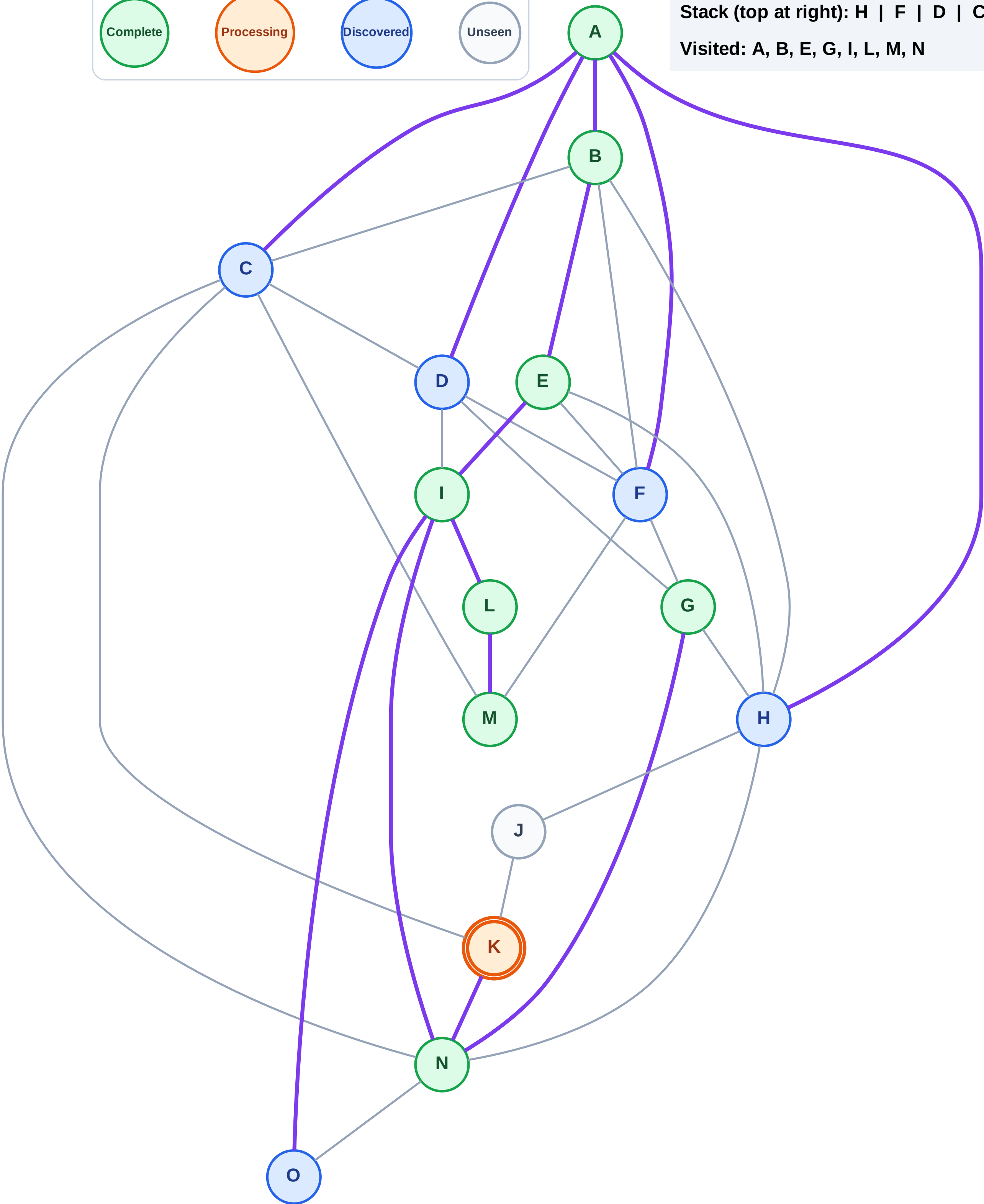
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O

Visited: A, B, E, G, I, L, M, N



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (5 hops): A → B → E → I → N → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O
Visited: A, B, E, G, I, K, L, M, N

